Neural Network

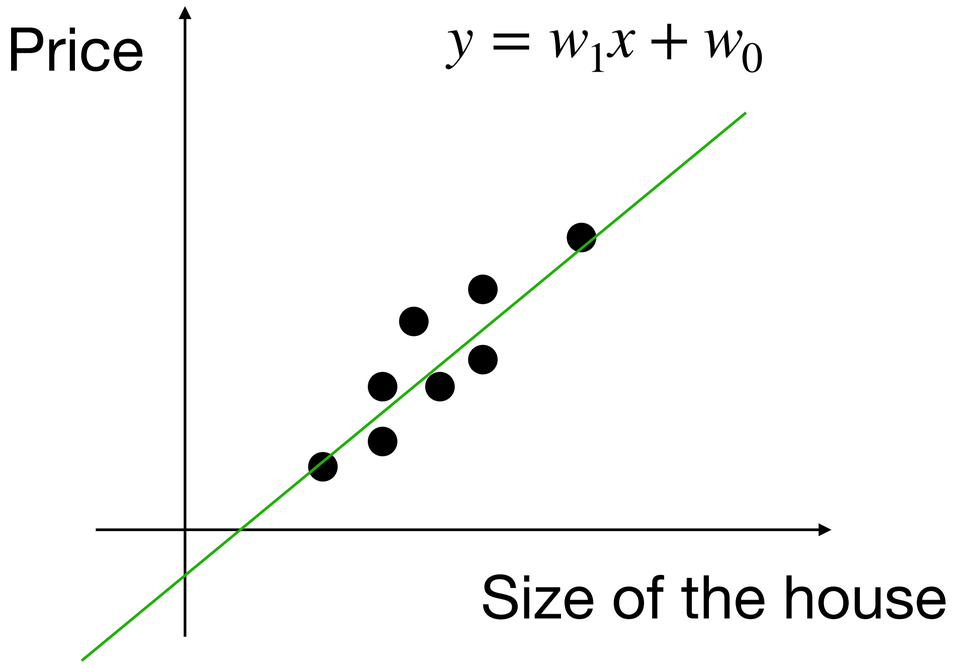
Outline of Today

1. Analysis of Boosting

2. Multilayer feedforward Neural Network

3. Training a neural network

Linear Regression Revisit



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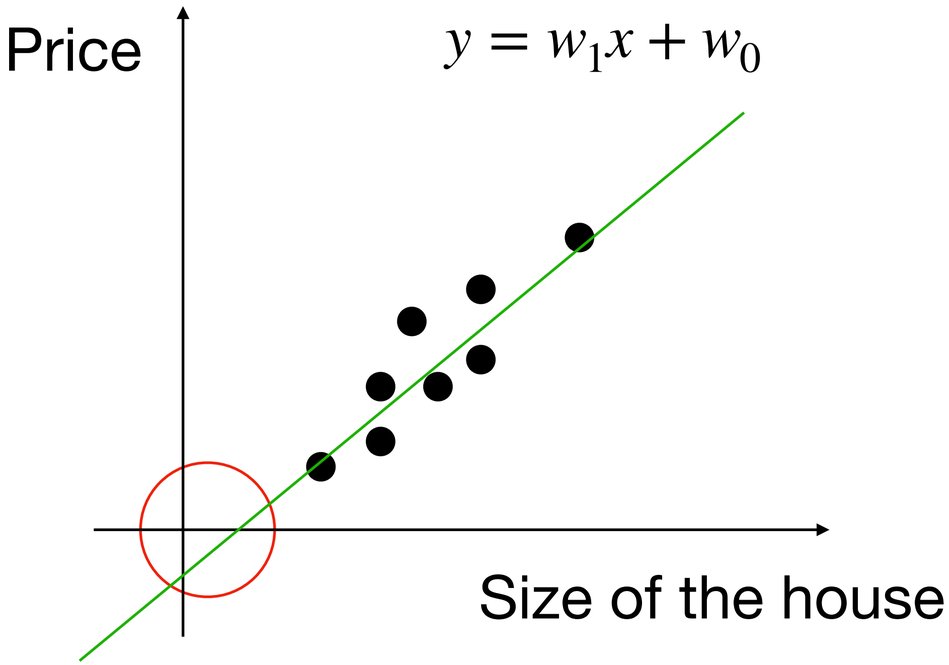

Negative part does not make too much sense

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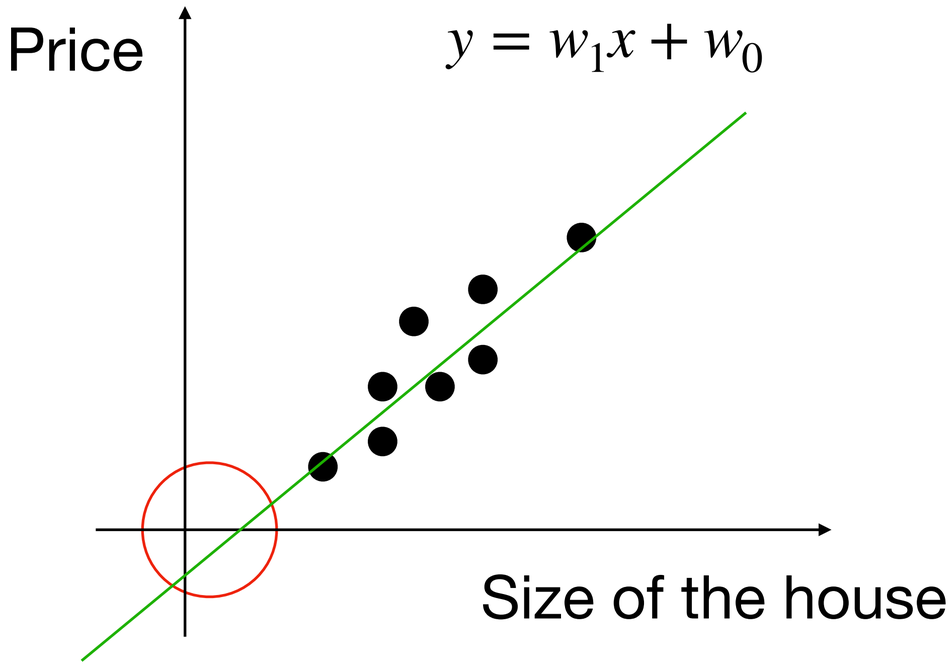

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We can fix this with a simple nonlinear function

$$y = \max\{w_1 x + w_0, 0\}$$

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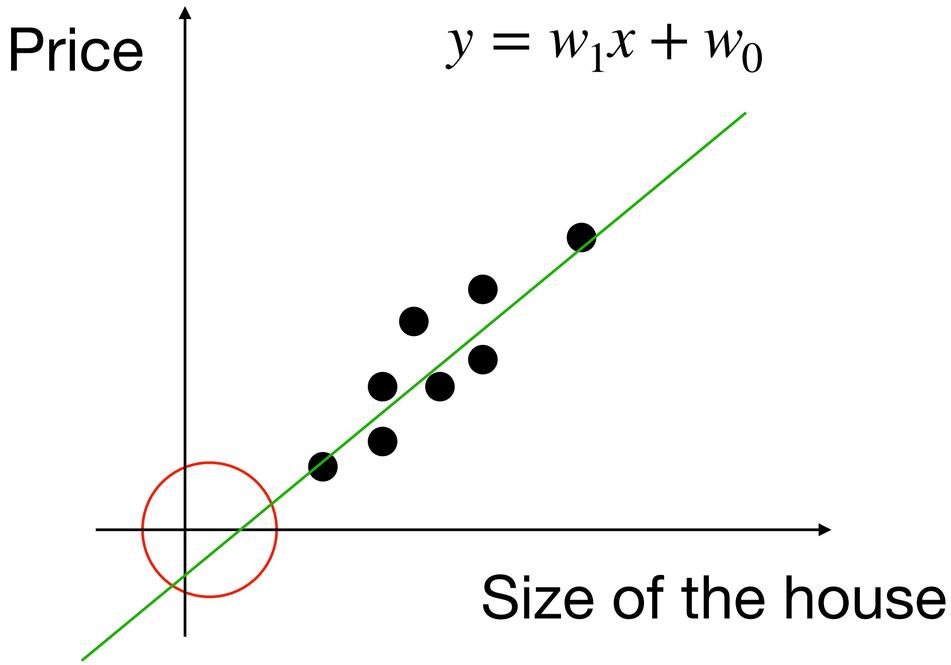

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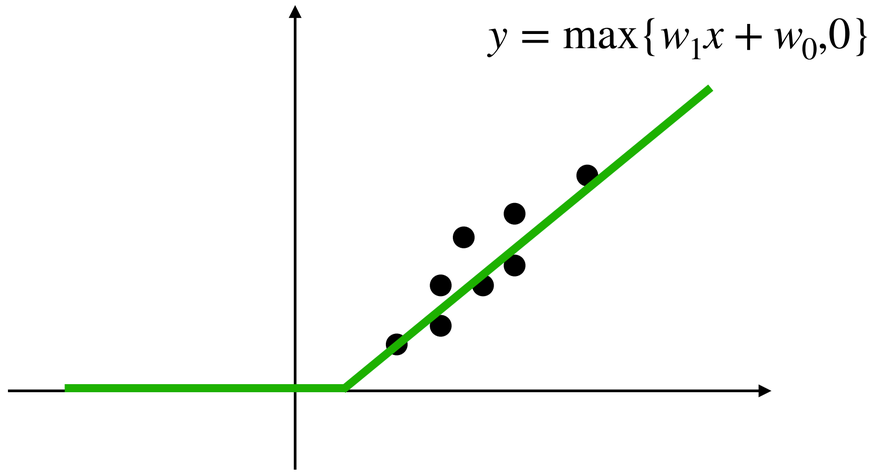

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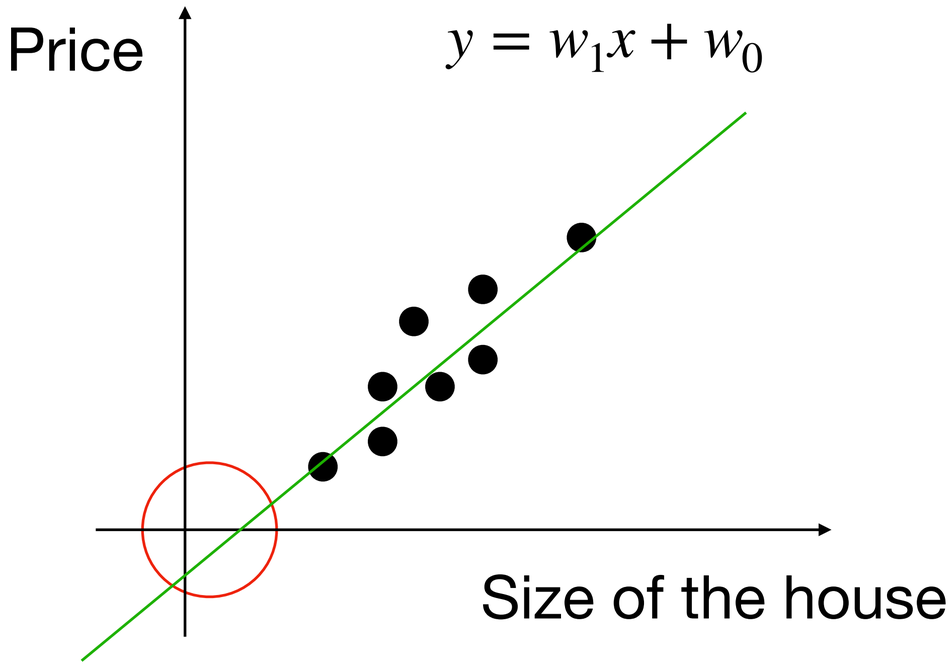

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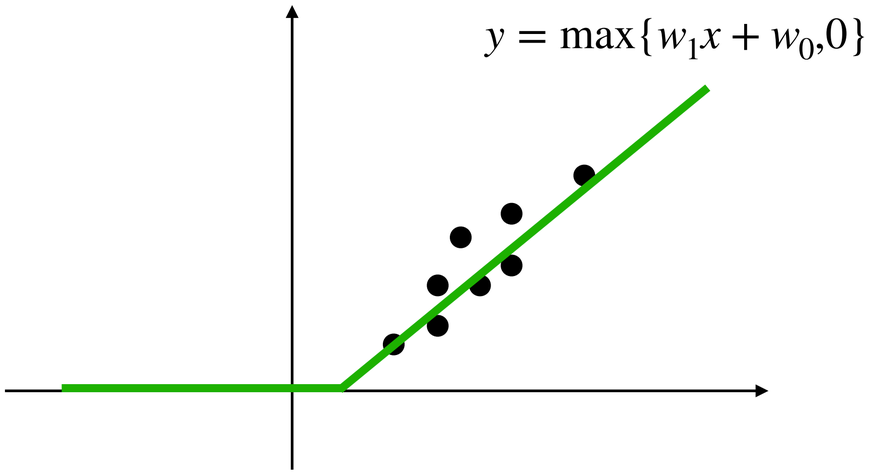
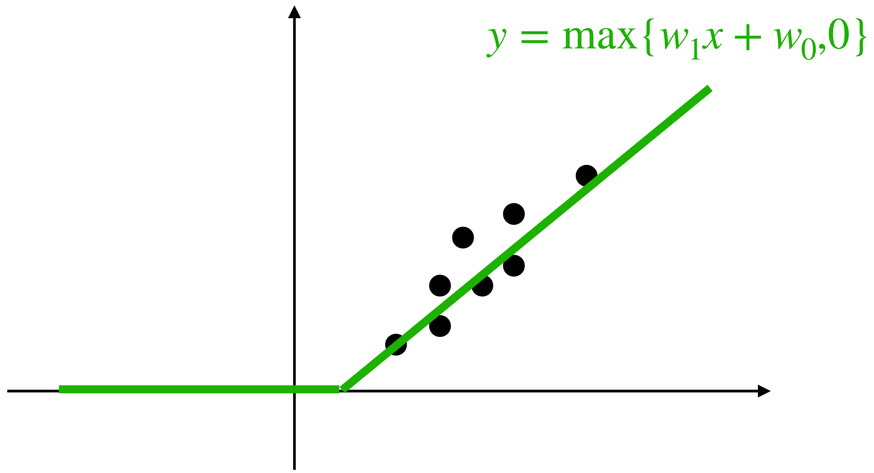


rectified linear unit (ReLU)

A single neuron network



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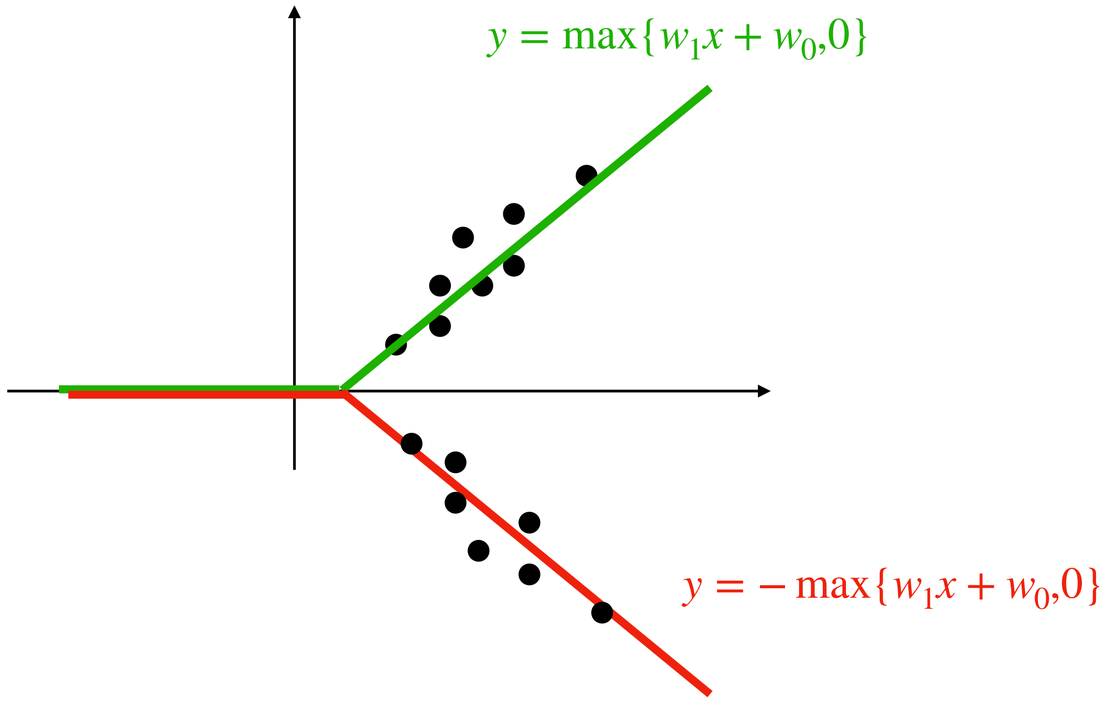

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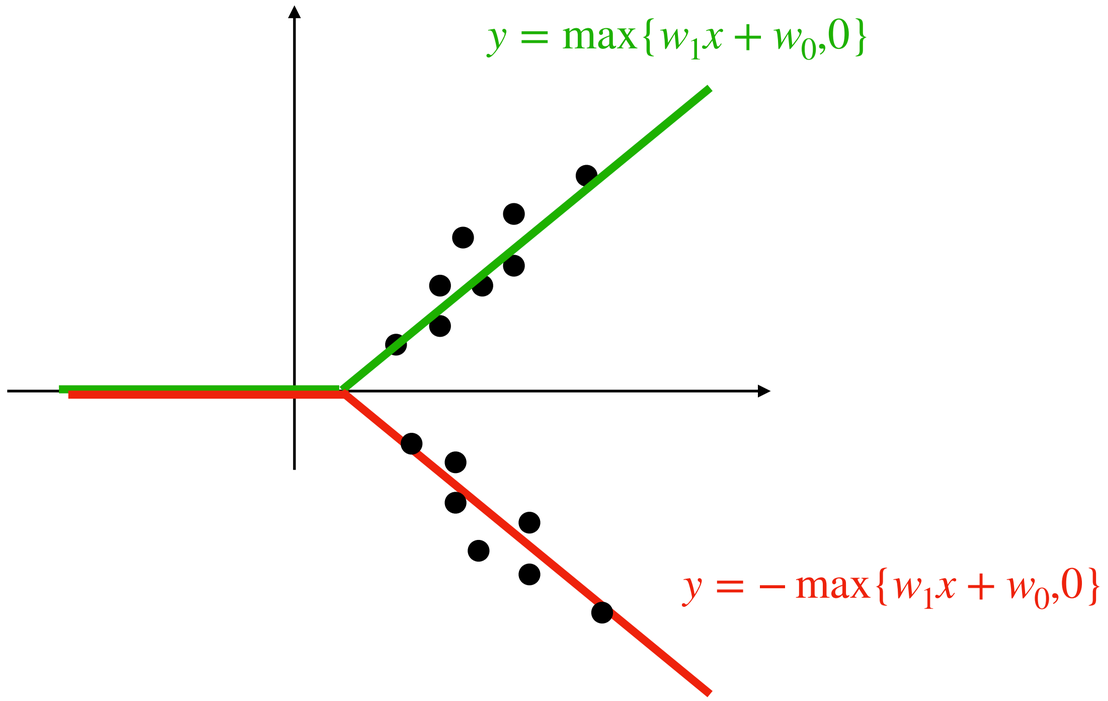

$$y = a \max\{w_1 x + w_0, 0\} + b$$

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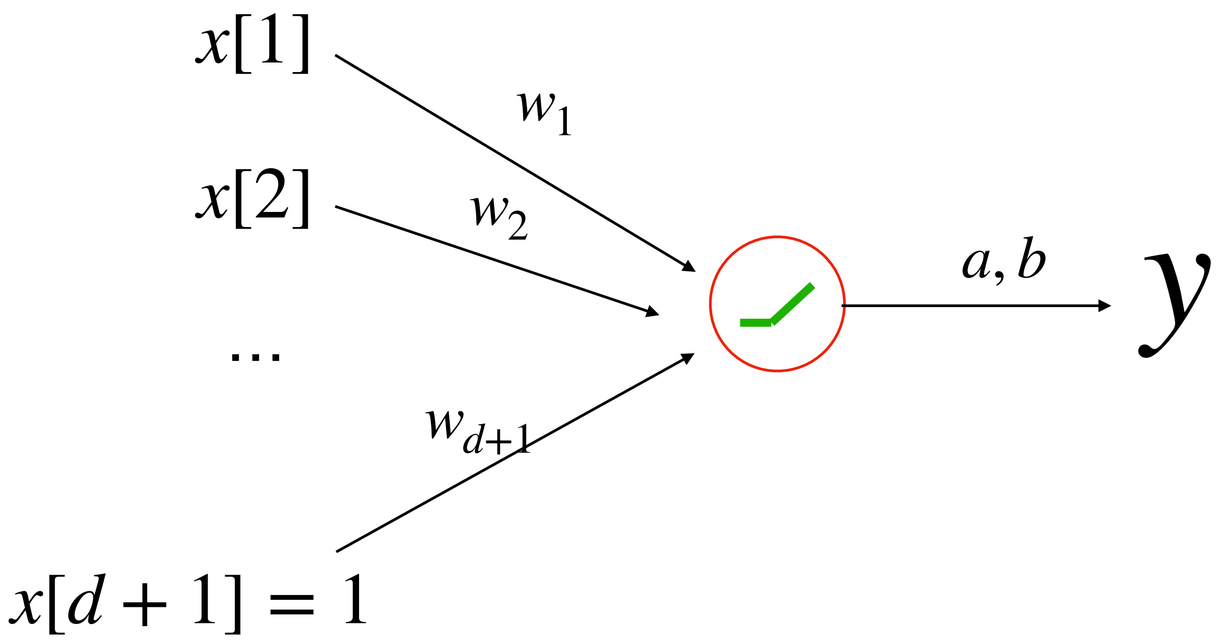

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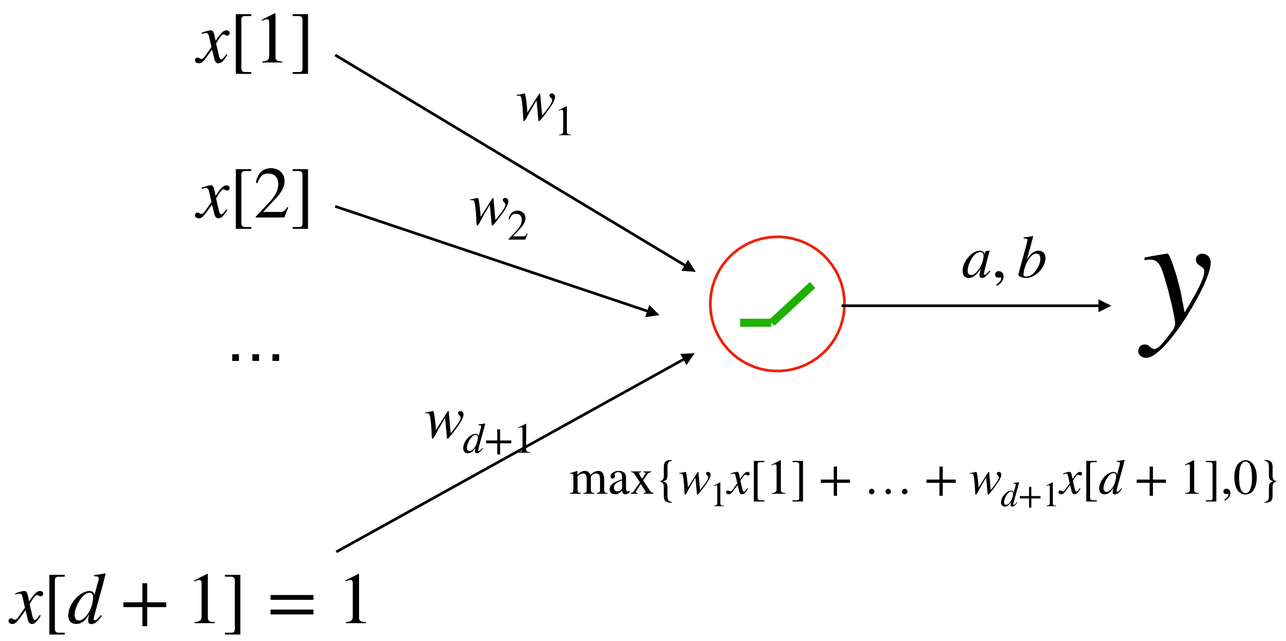

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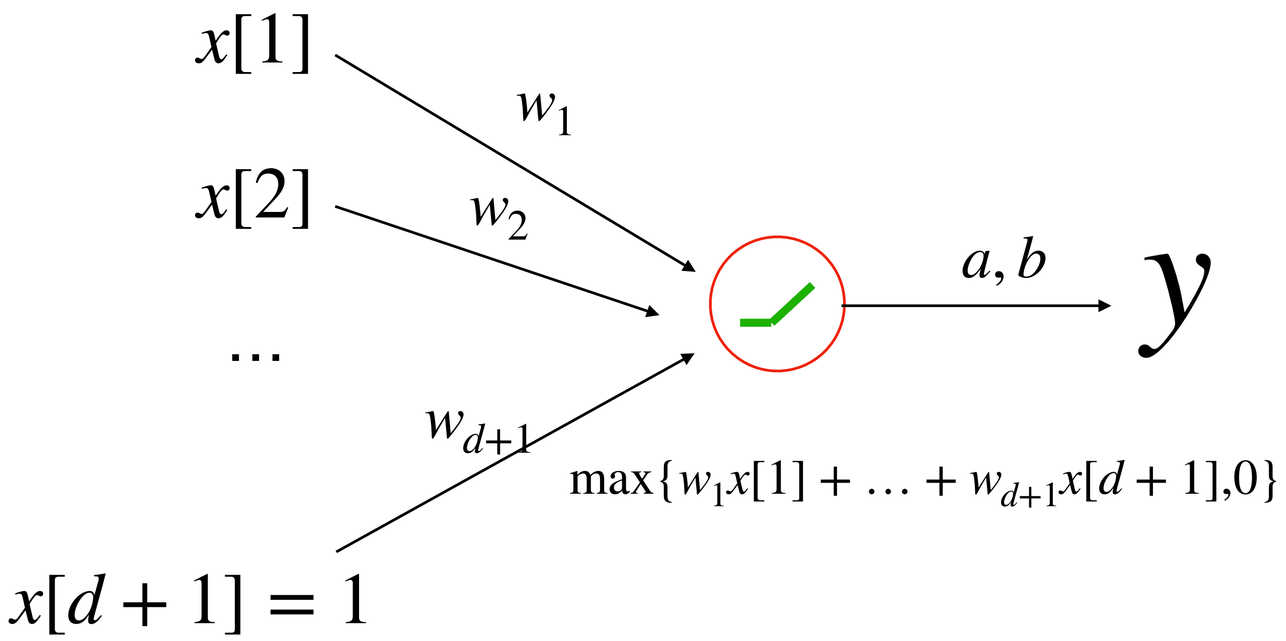
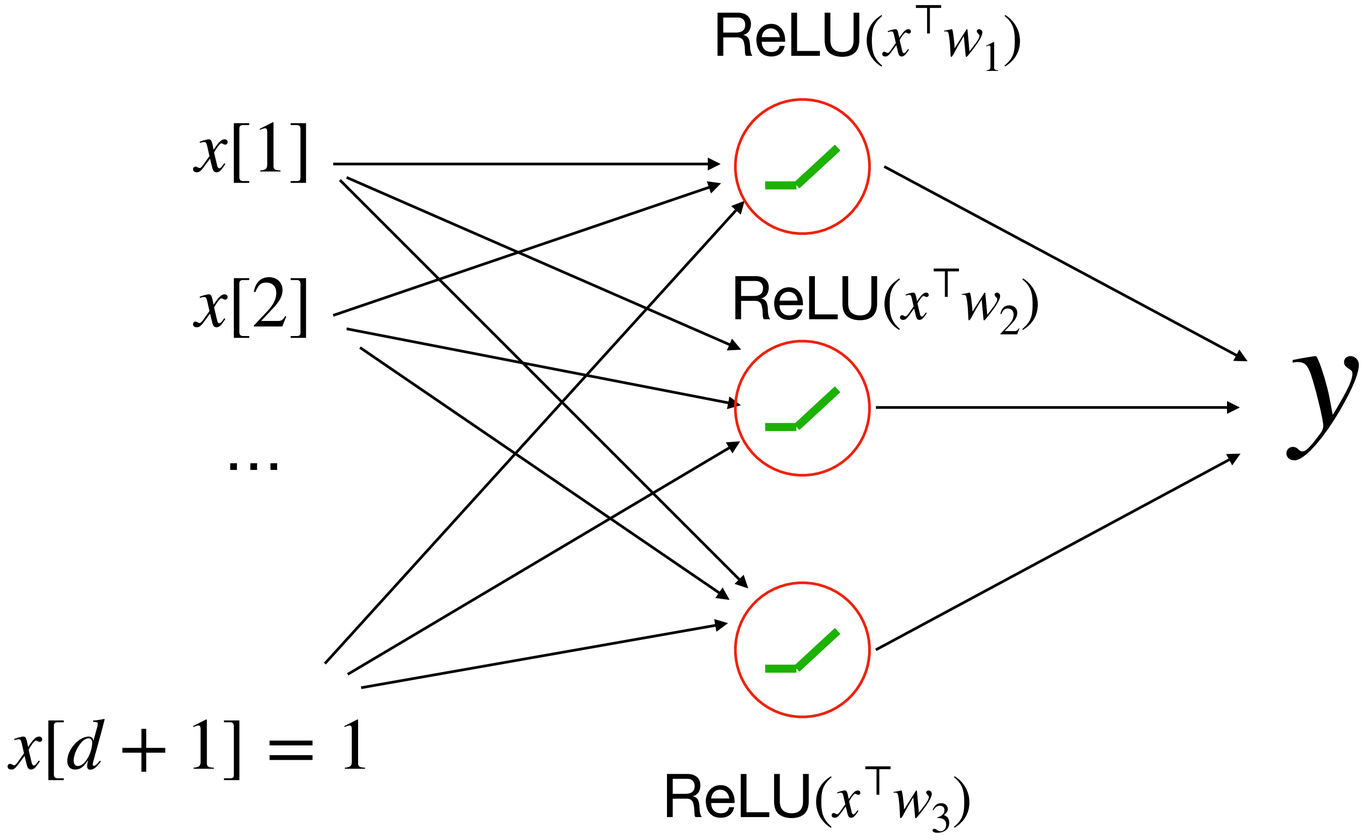


$$y = a\mathrm{ReLU}(w^\top x) + b$$

Let us stack multiple neurons together



$$y = \sum_{i=1}^{3} a_i \mathrm{ReLU}(x^\top w_i) + b$$

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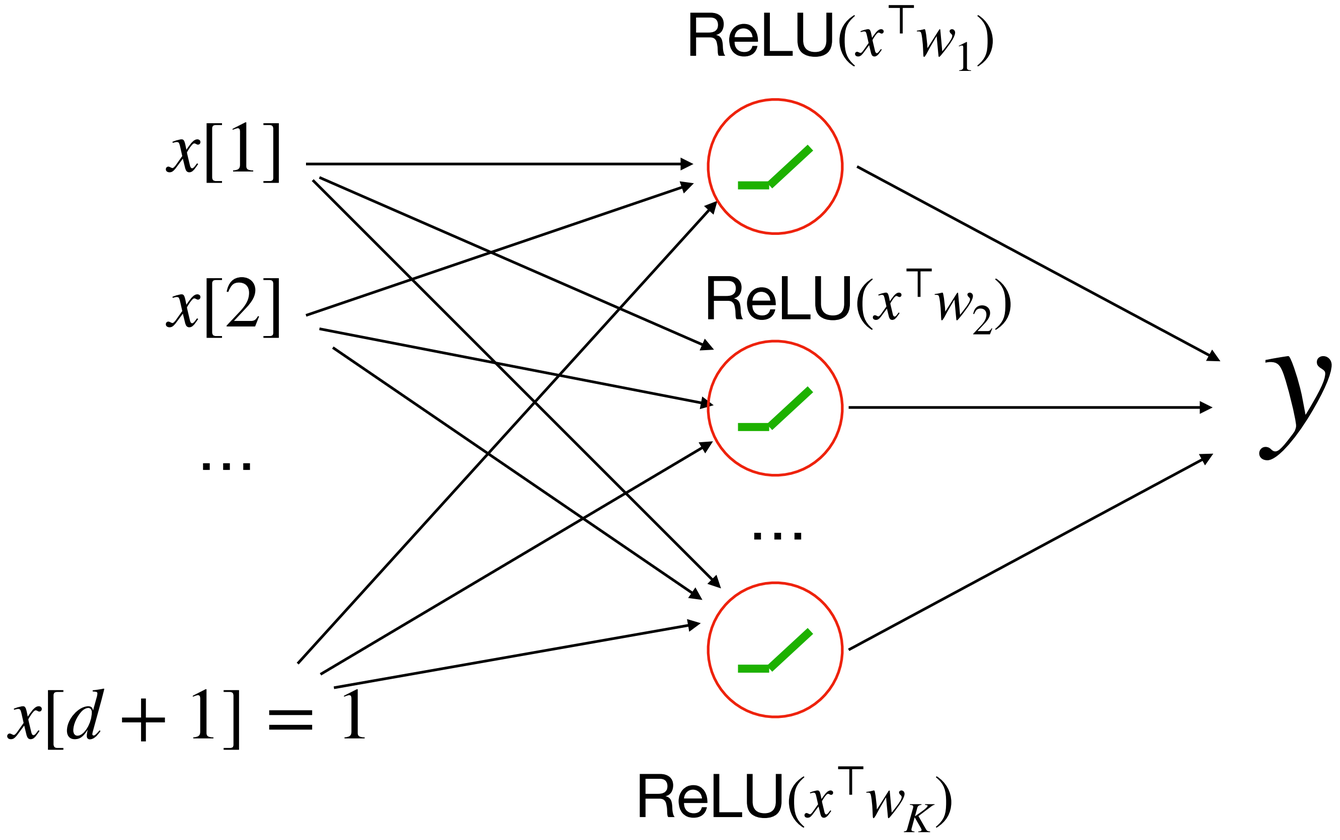

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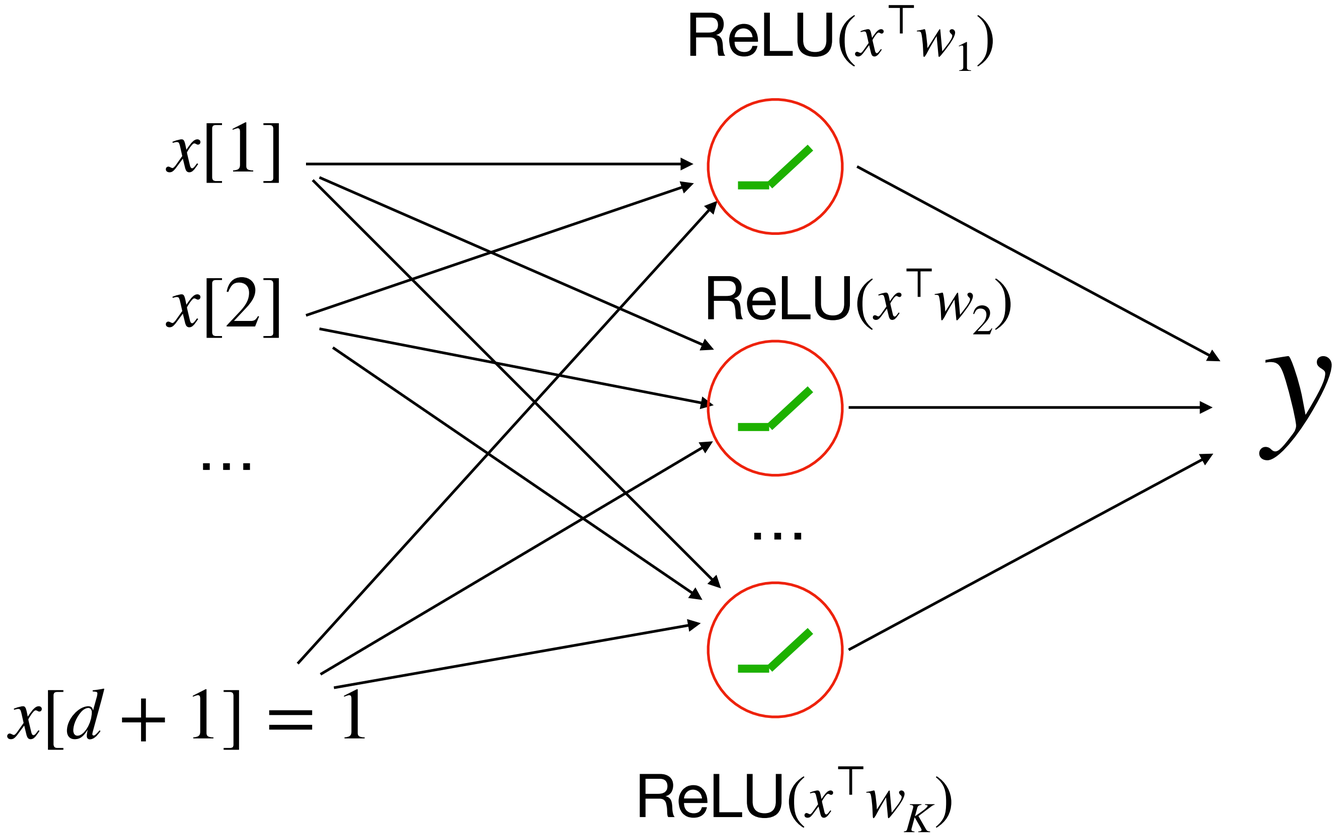

Vectorized form:

Define $W = \begin{bmatrix} (w_1)^\top \\ \cdots \\ (w_K)^\top \end{bmatrix} \in \mathbb{R}^{K \times d}$

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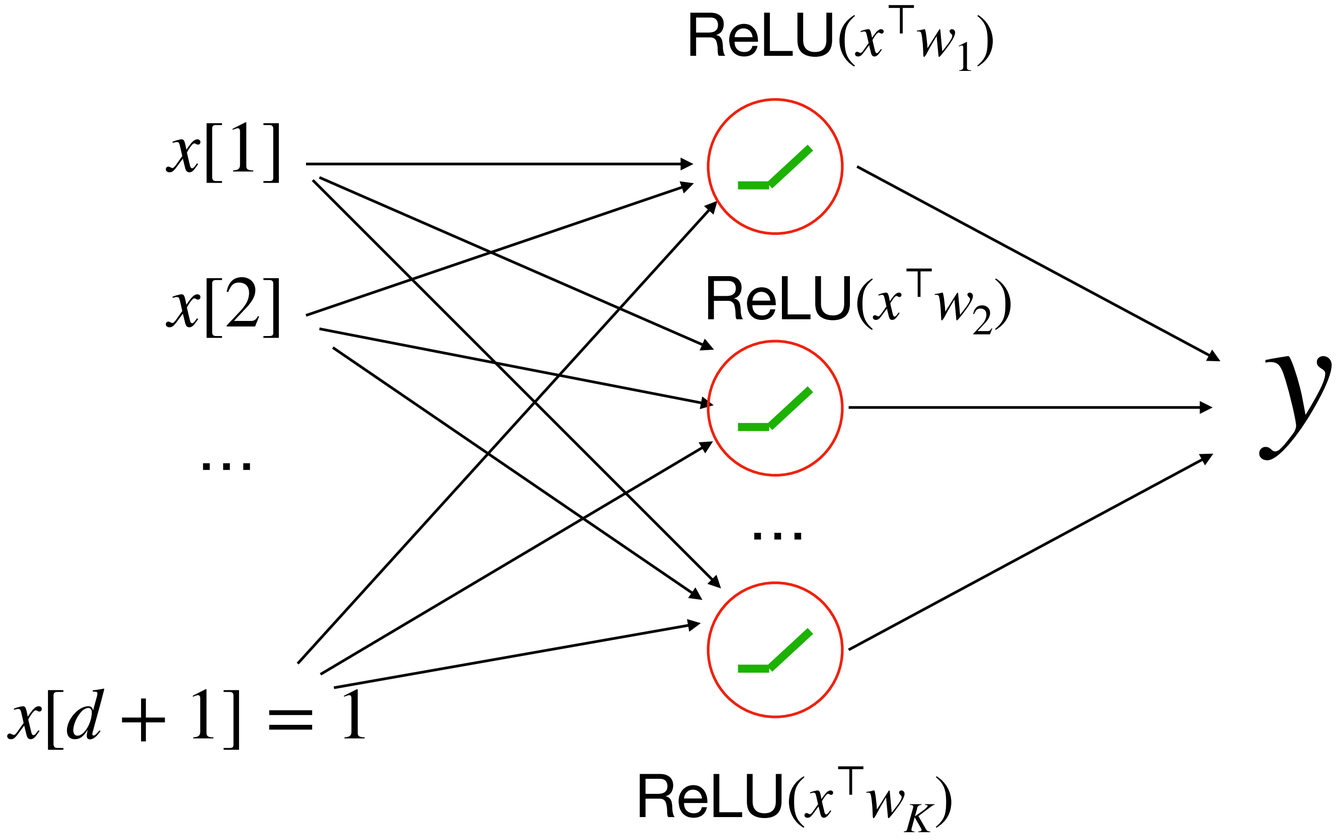

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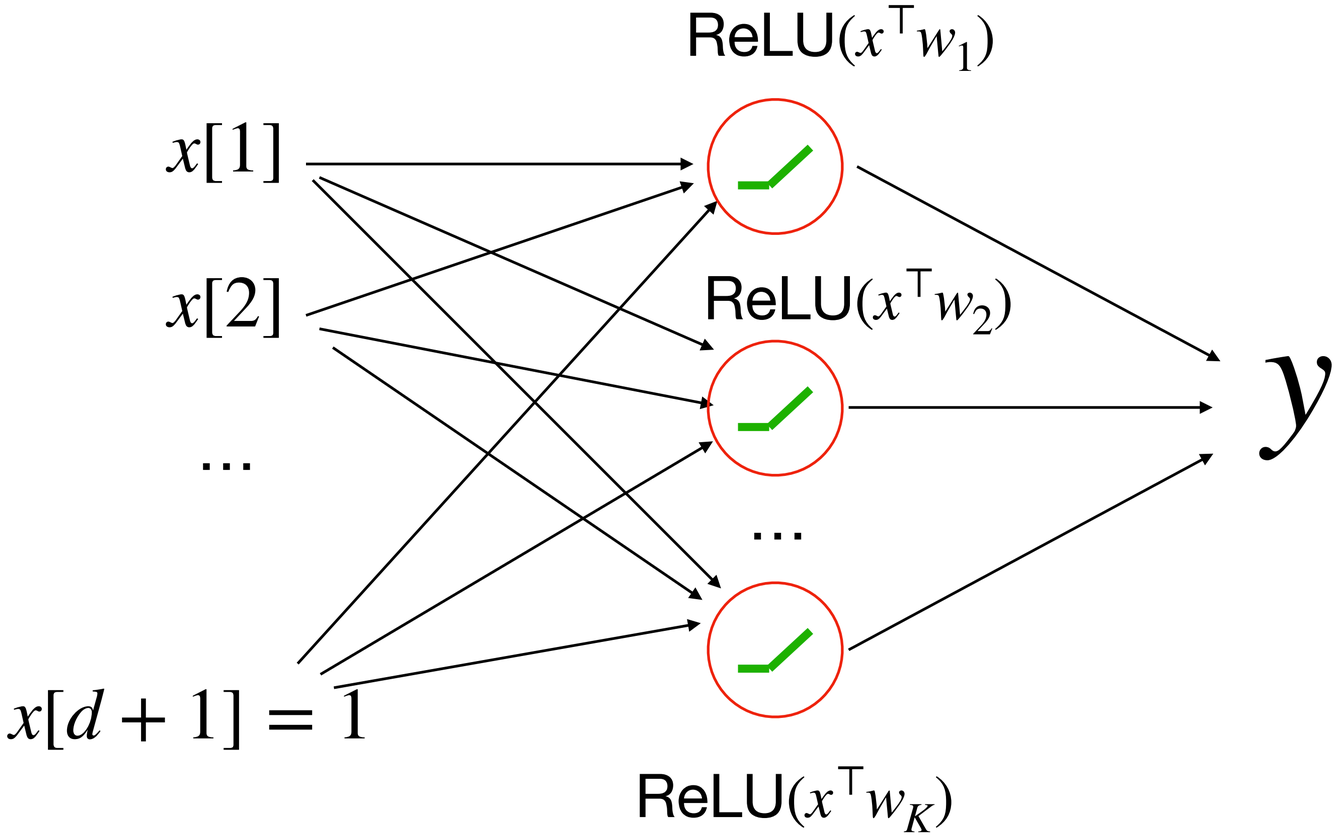

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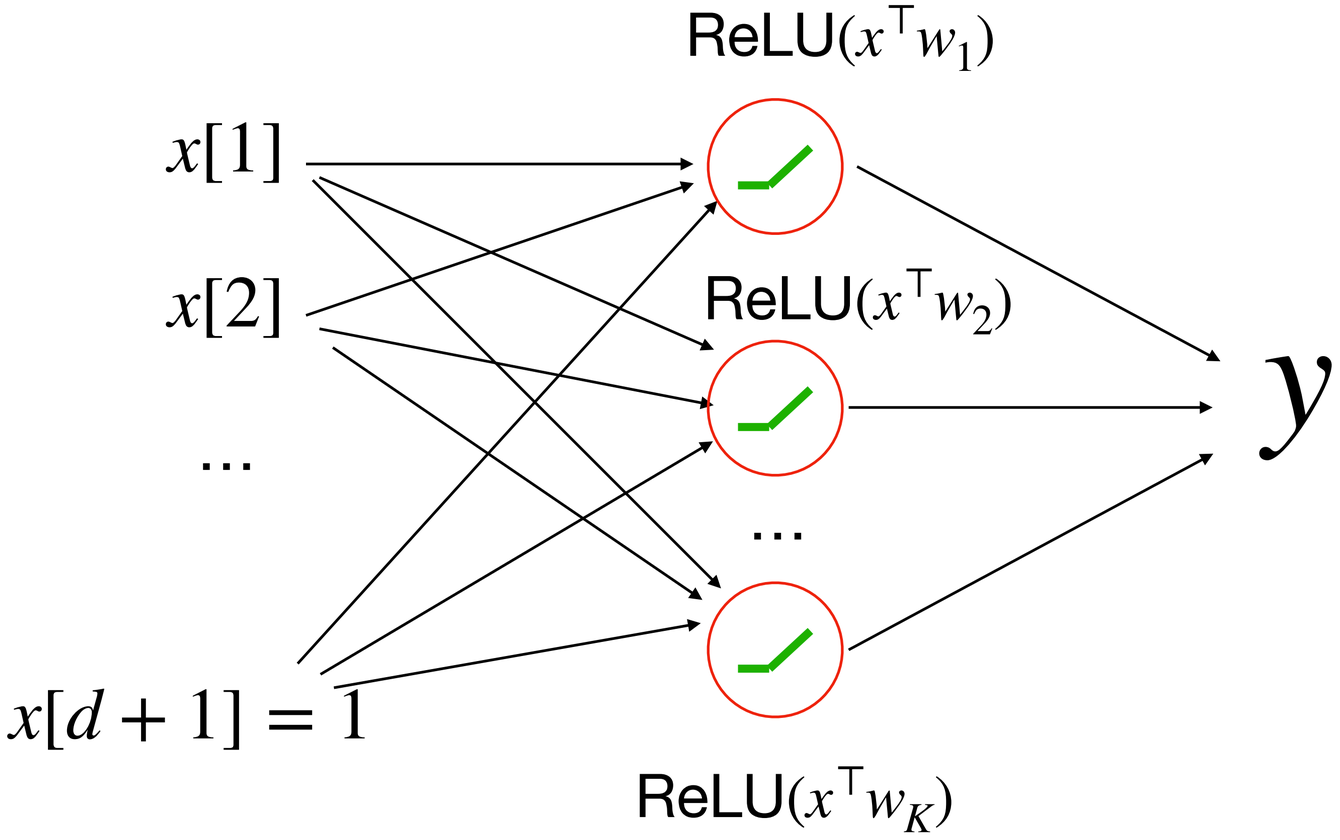

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Learnable feature $\phi(x)$

What does a neural network approximate

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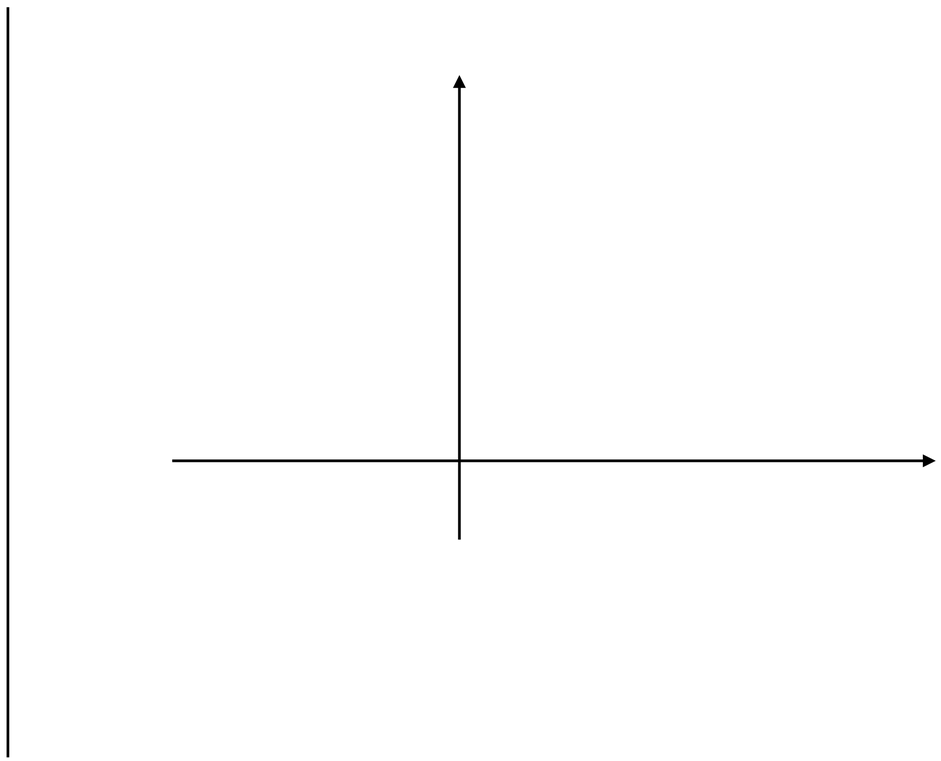

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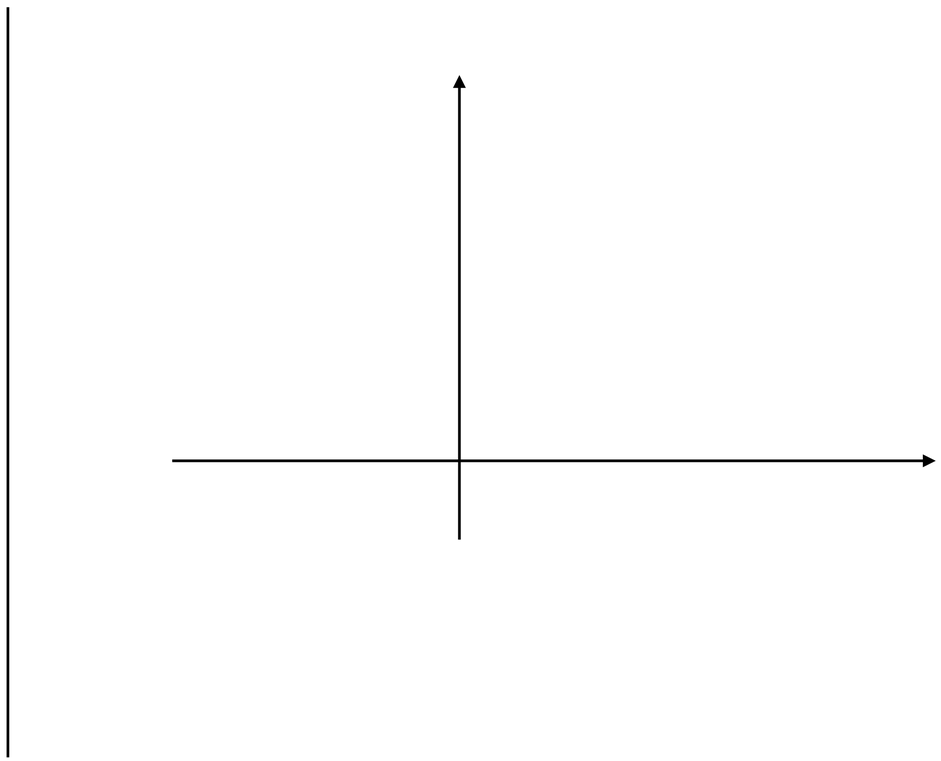

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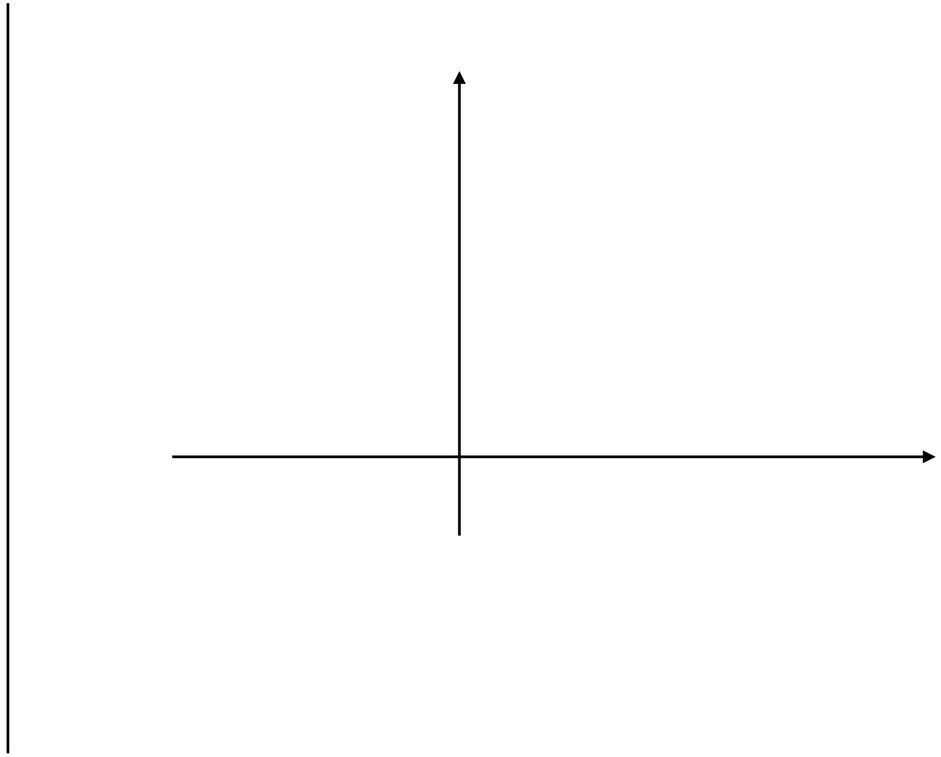

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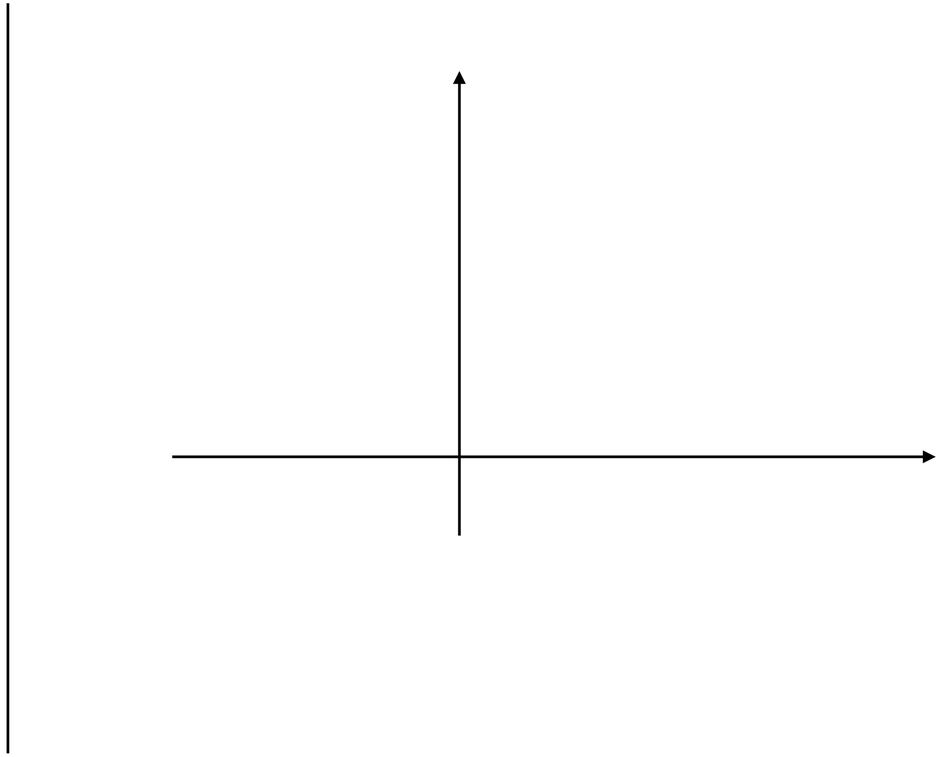

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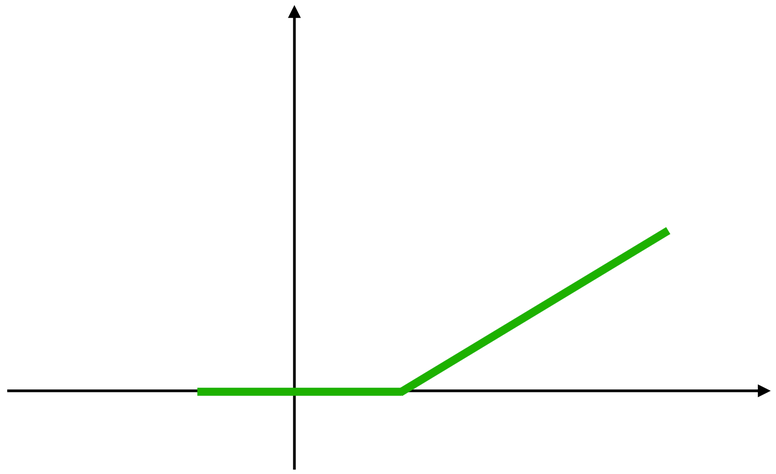

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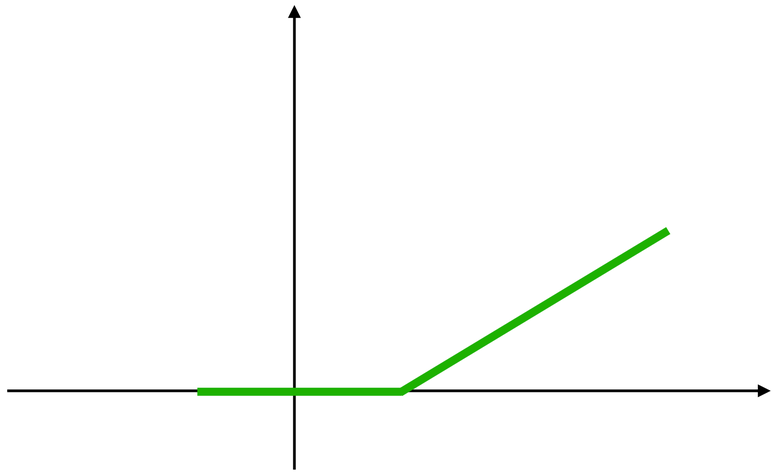

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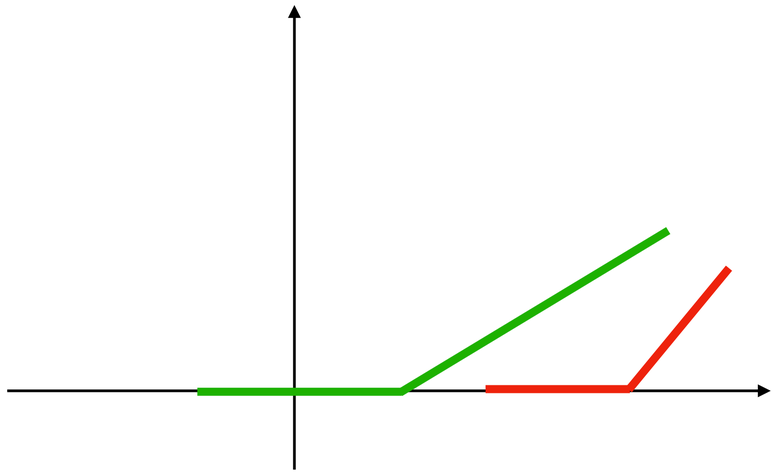

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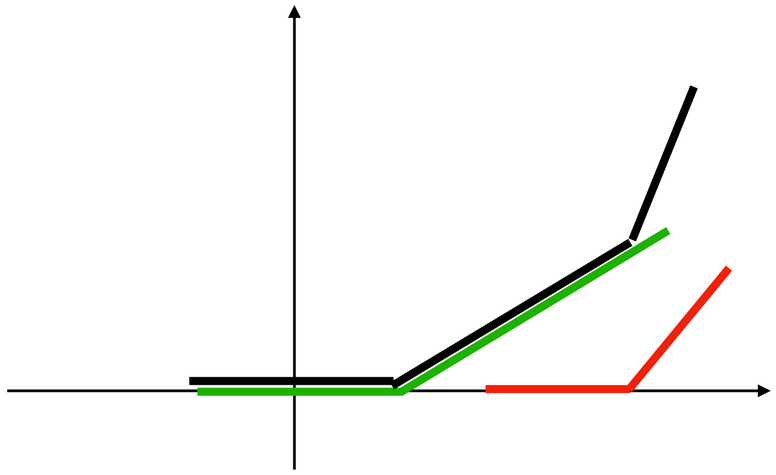

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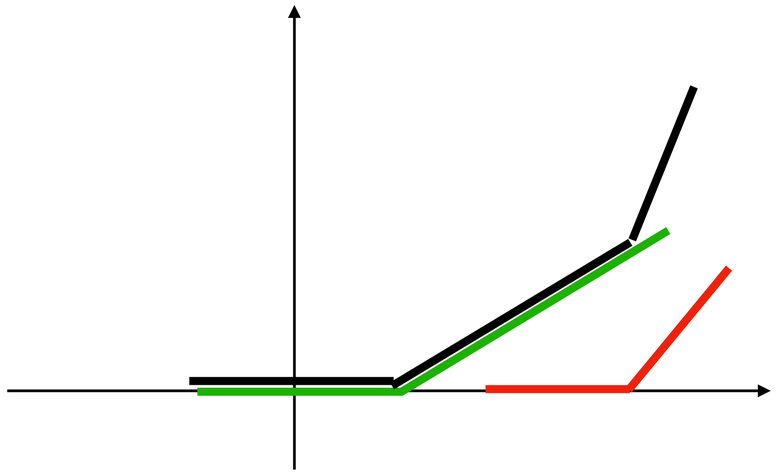

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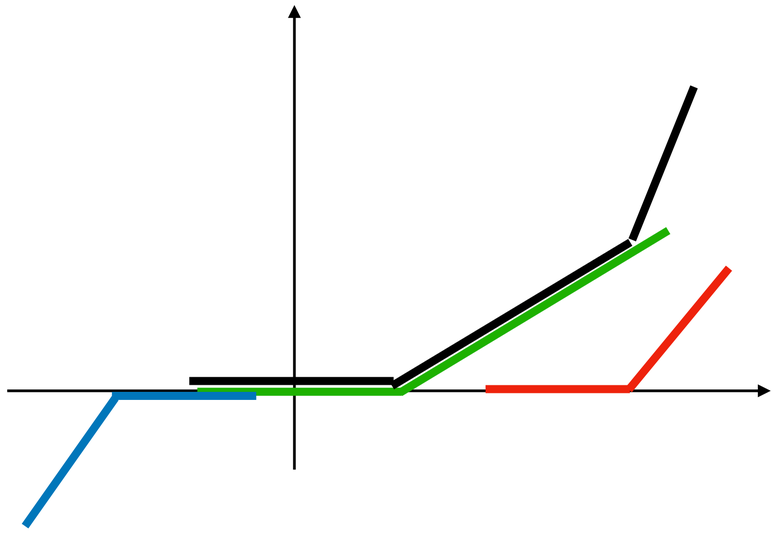

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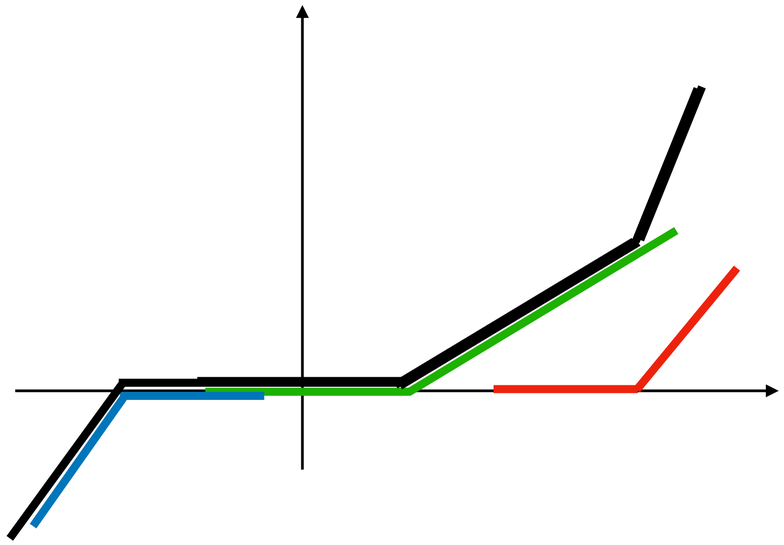

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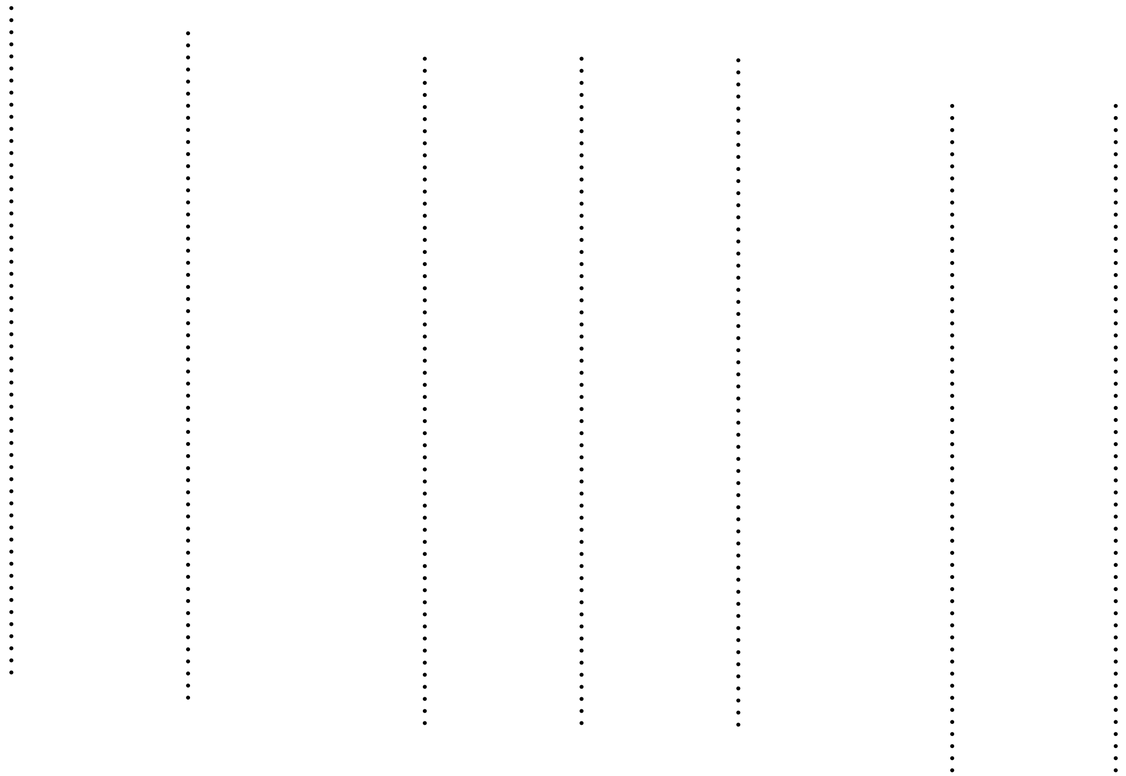
Claim: a wide enough one layer NN can approximate any smooth functions



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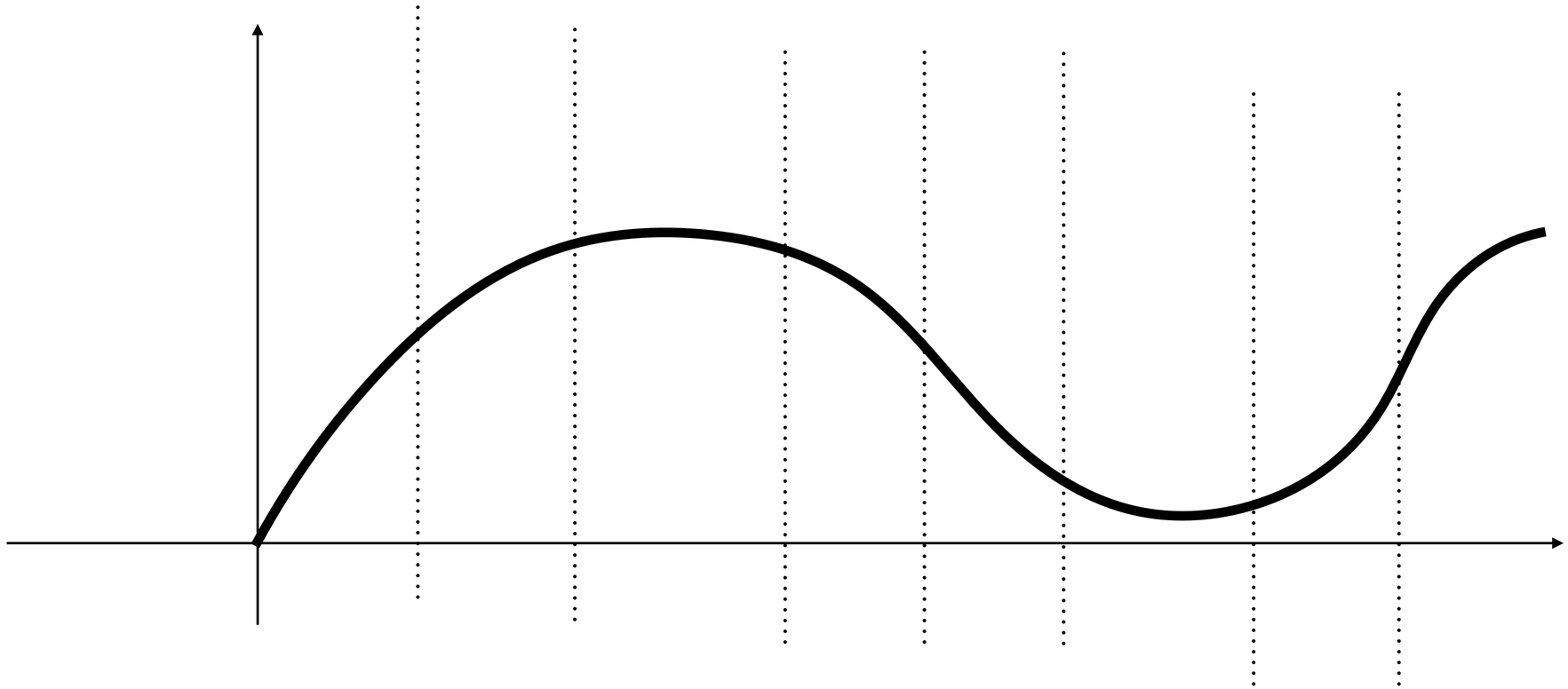

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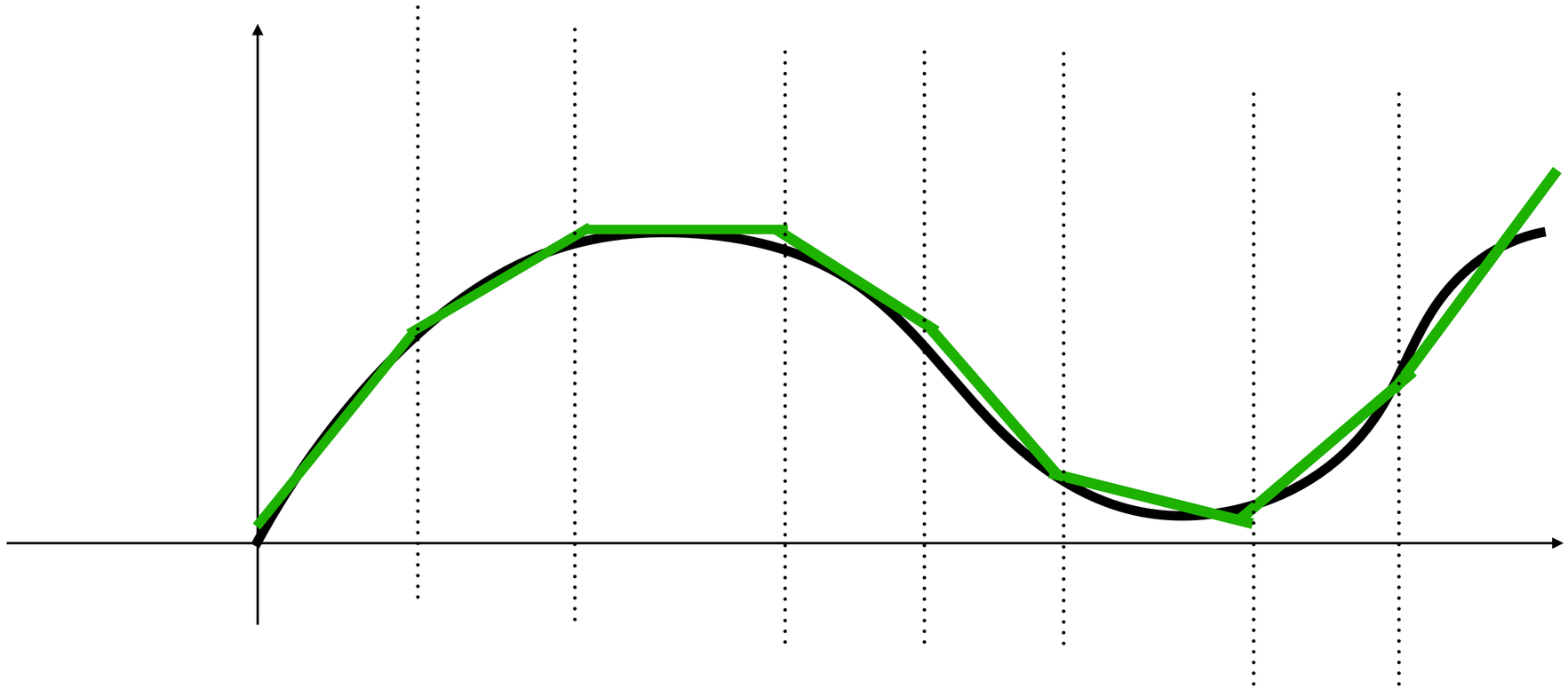

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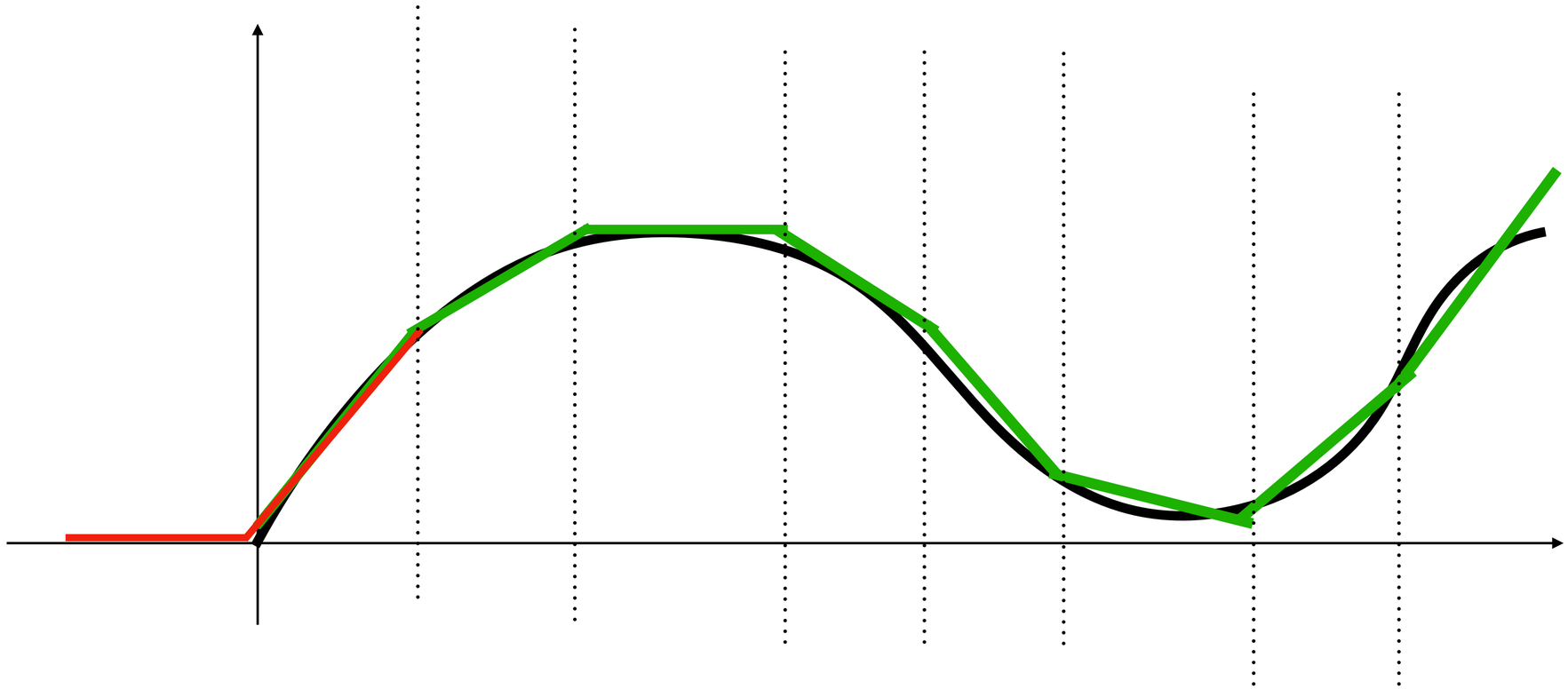

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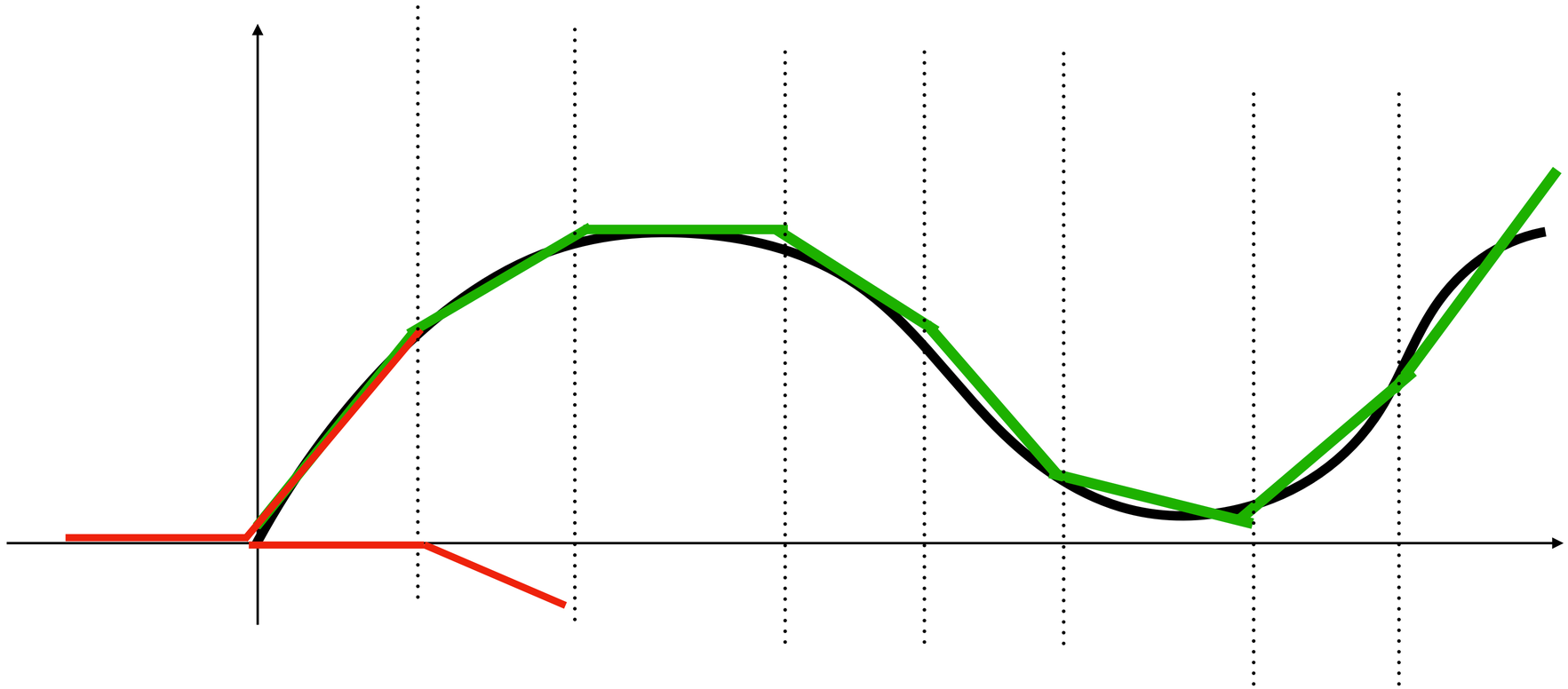

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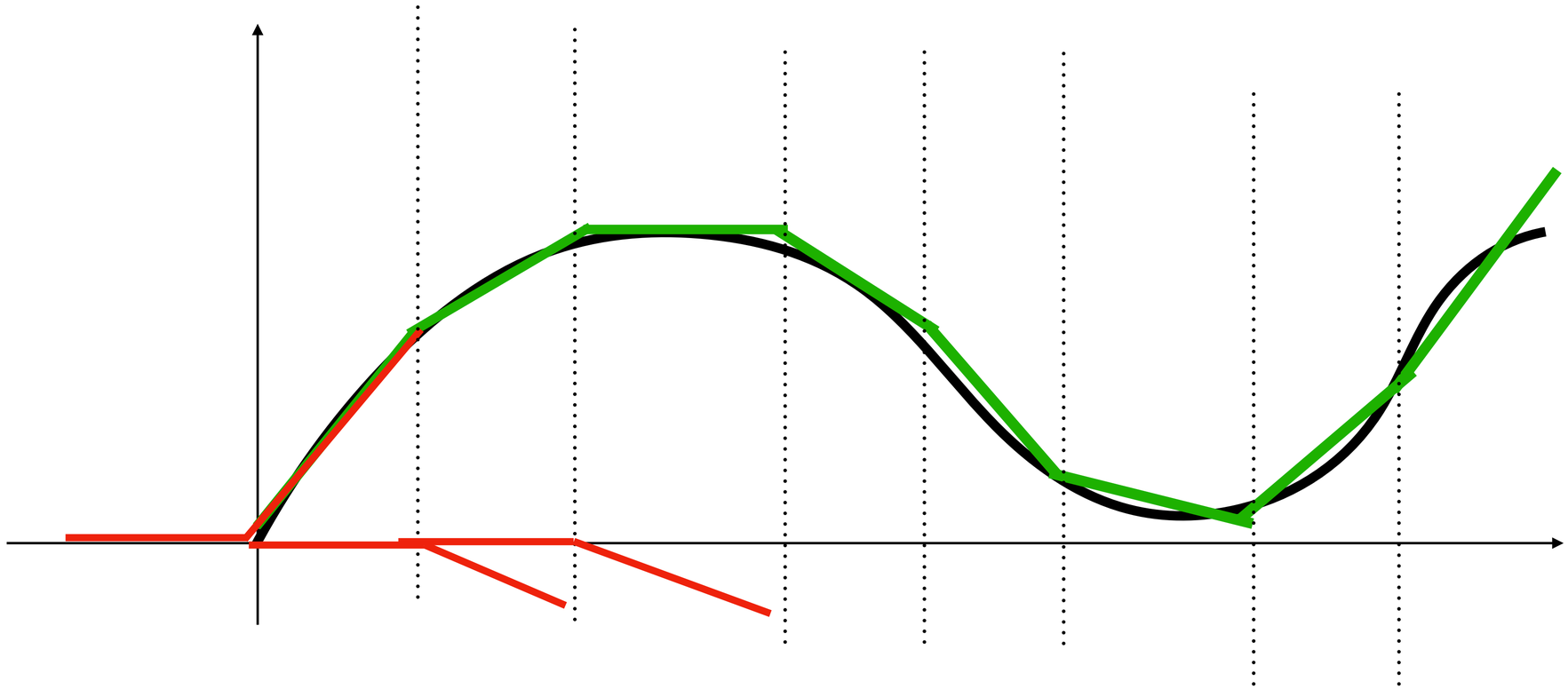

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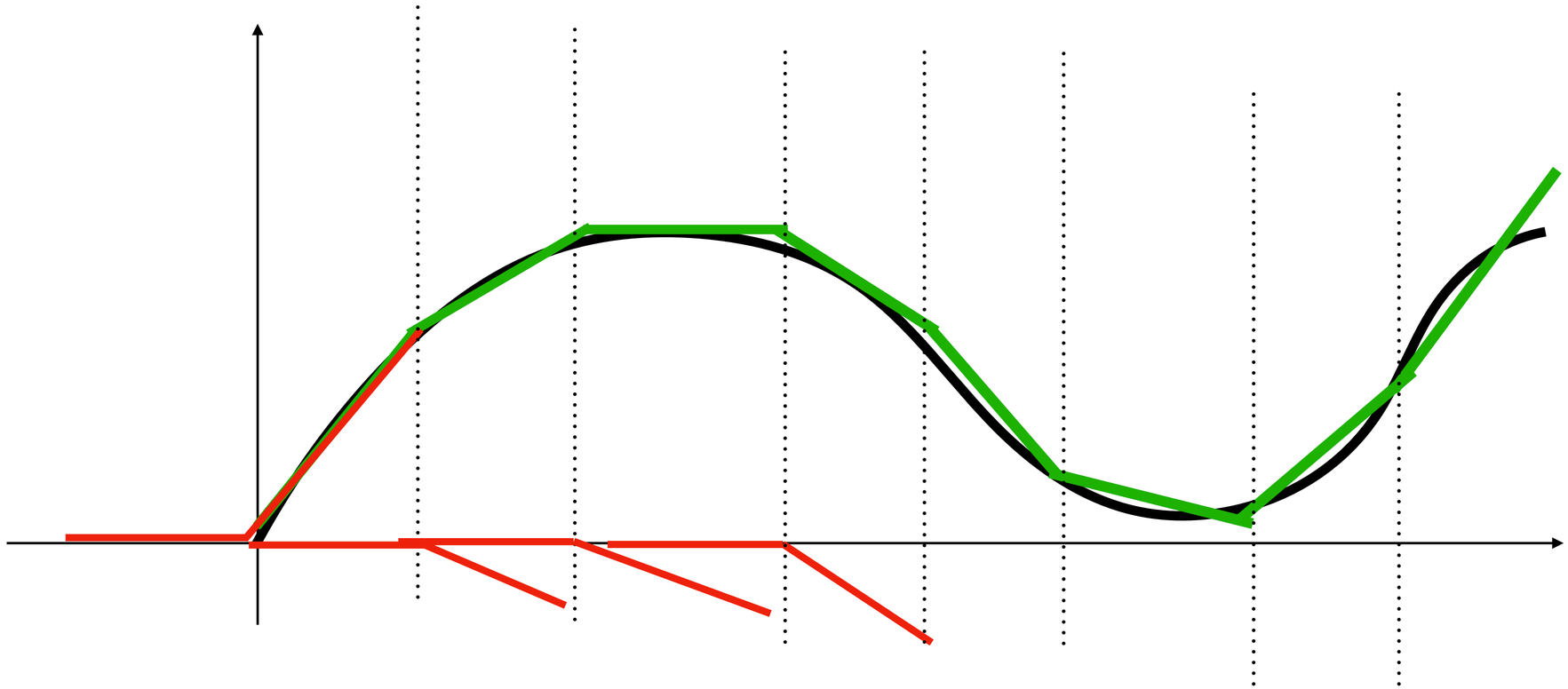
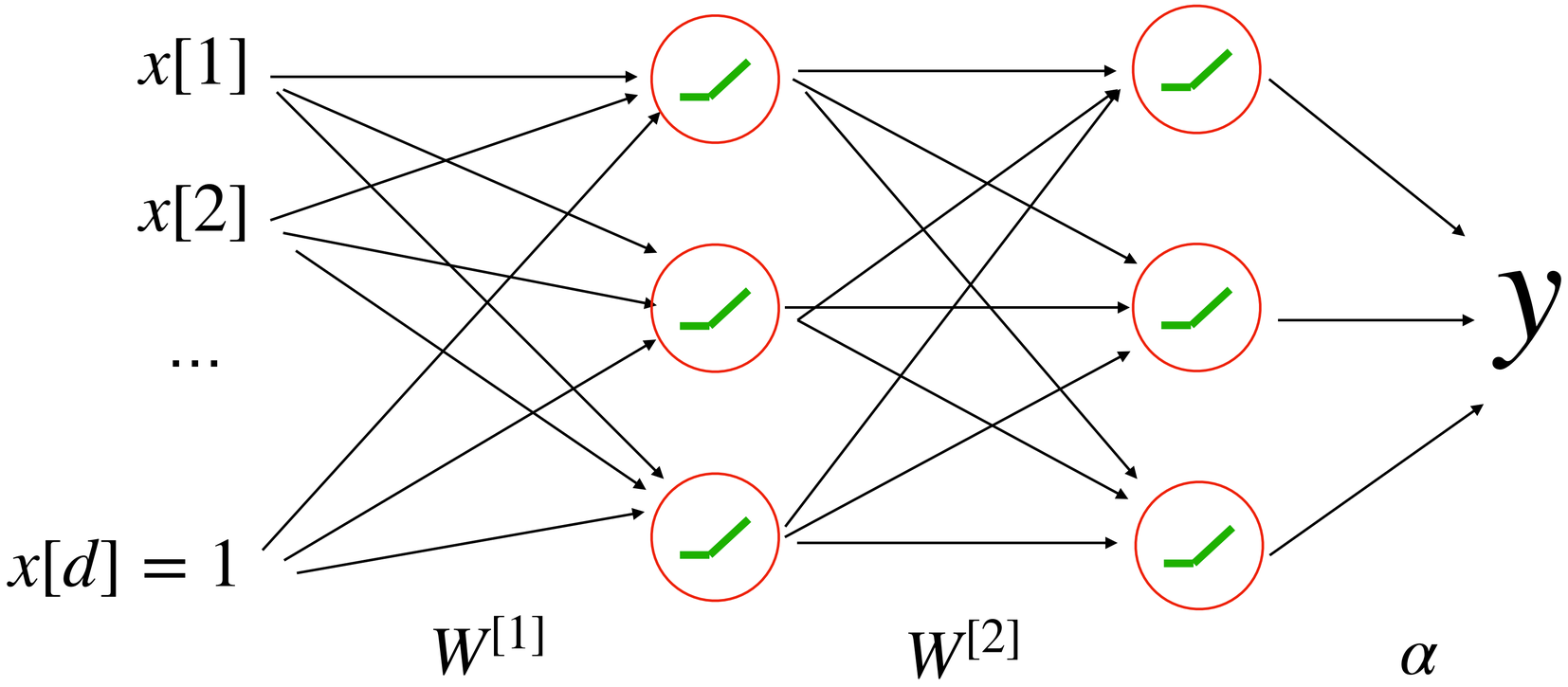


A multi-layer fully connected neural network



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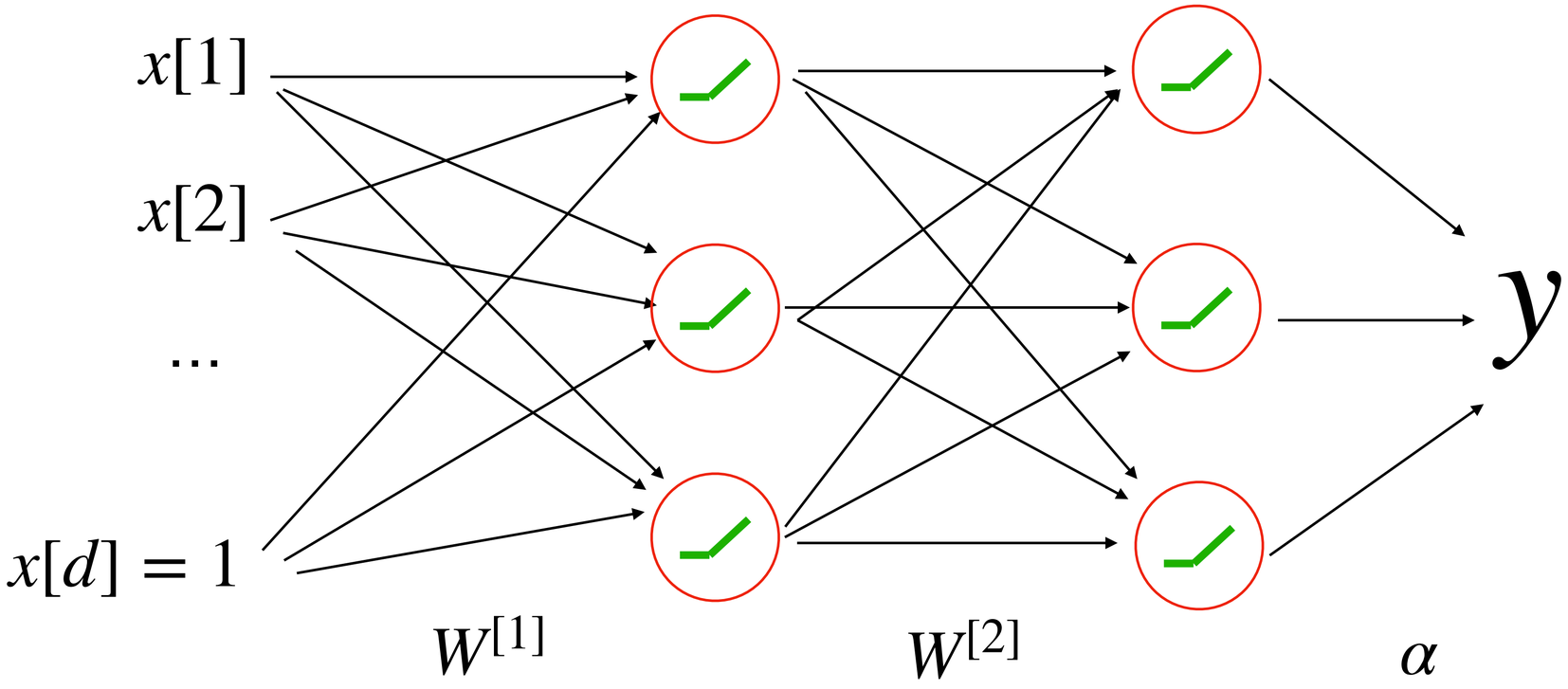
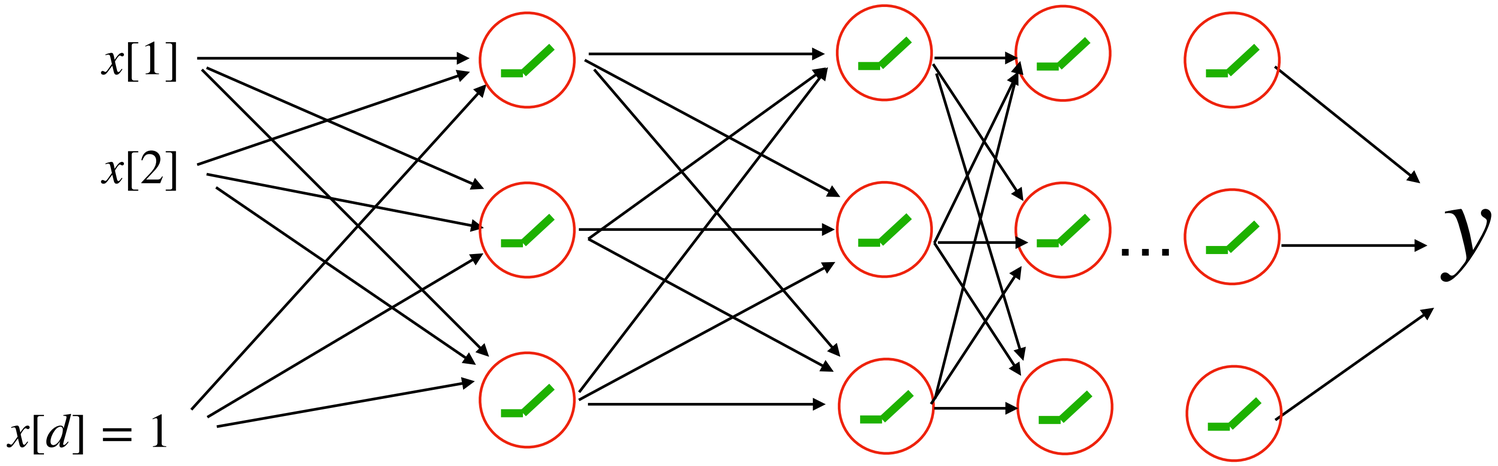


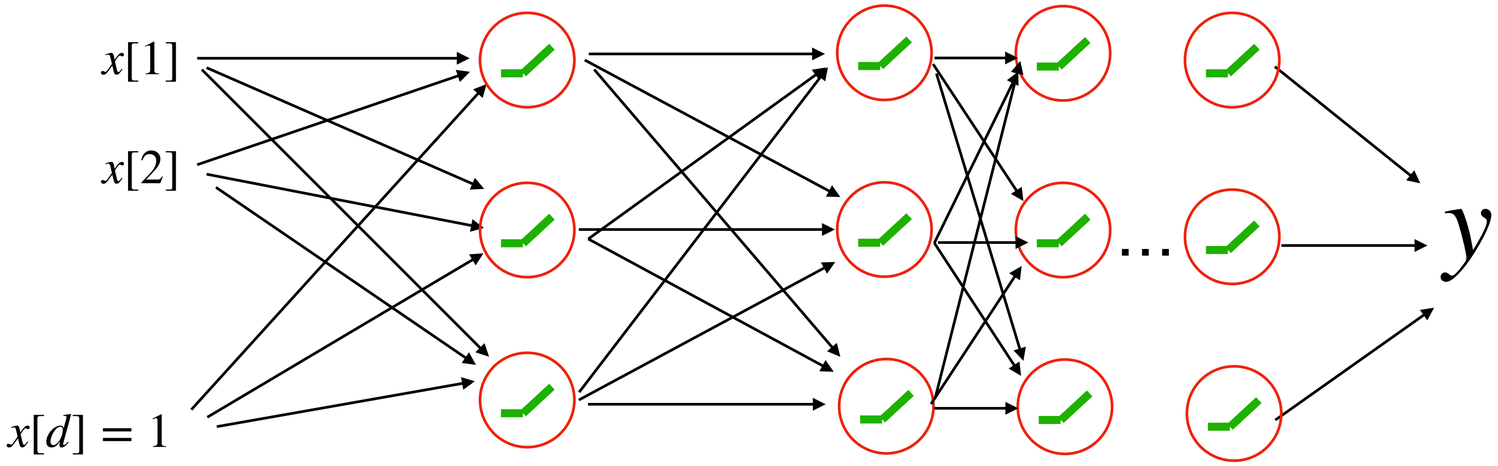
$$y = \alpha^{\top}\text{ReLU}\left(W^{[2]}\text{ReLU}\left(W^{[1]}x\right)\right) + b$$

A multi-layer fully connected neural network



A multi-layer fully connected neural network

Define it by a forward pass:



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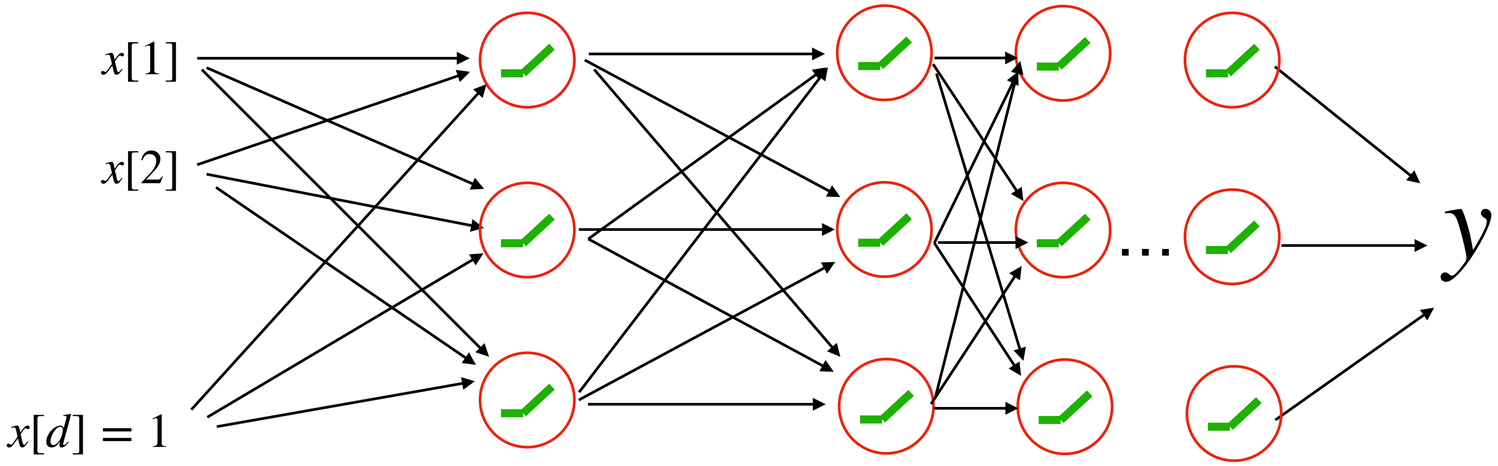

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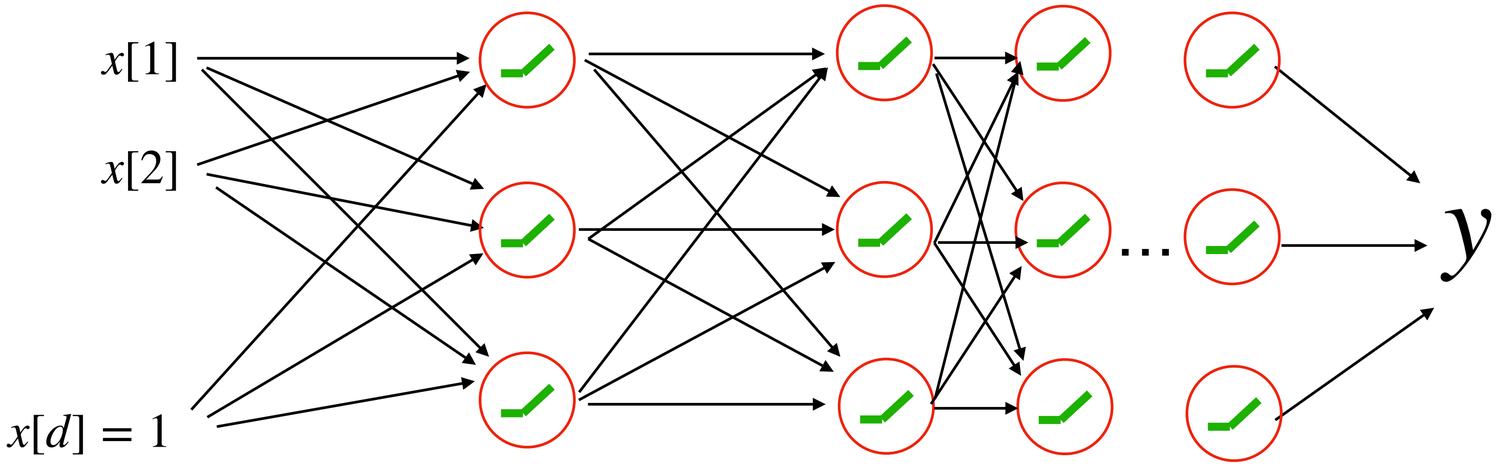

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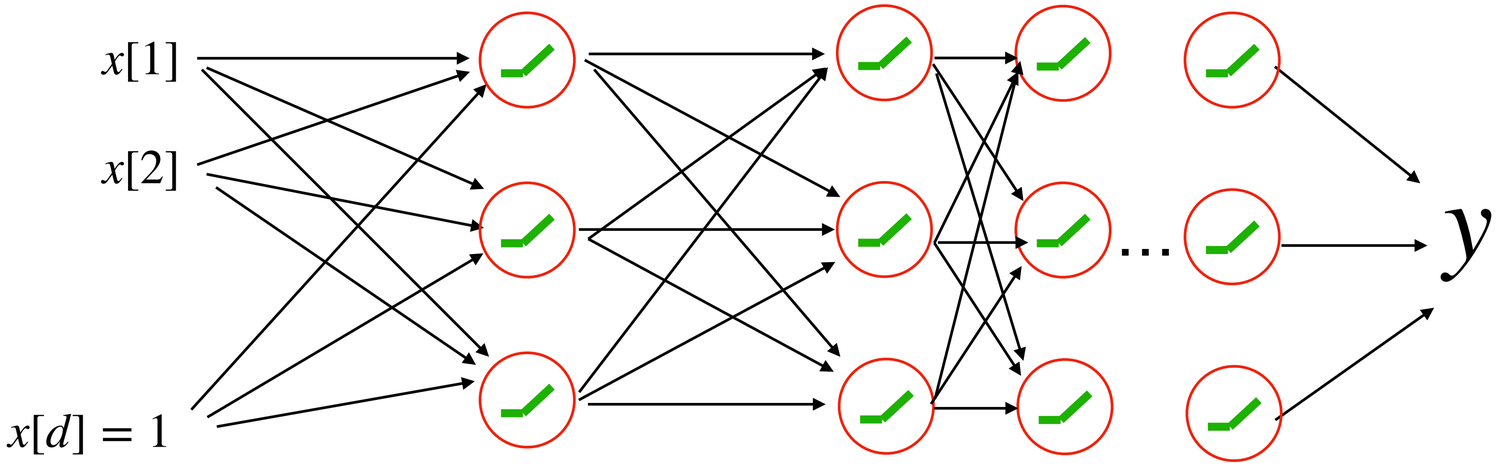
A multi-layer fully connected neural network



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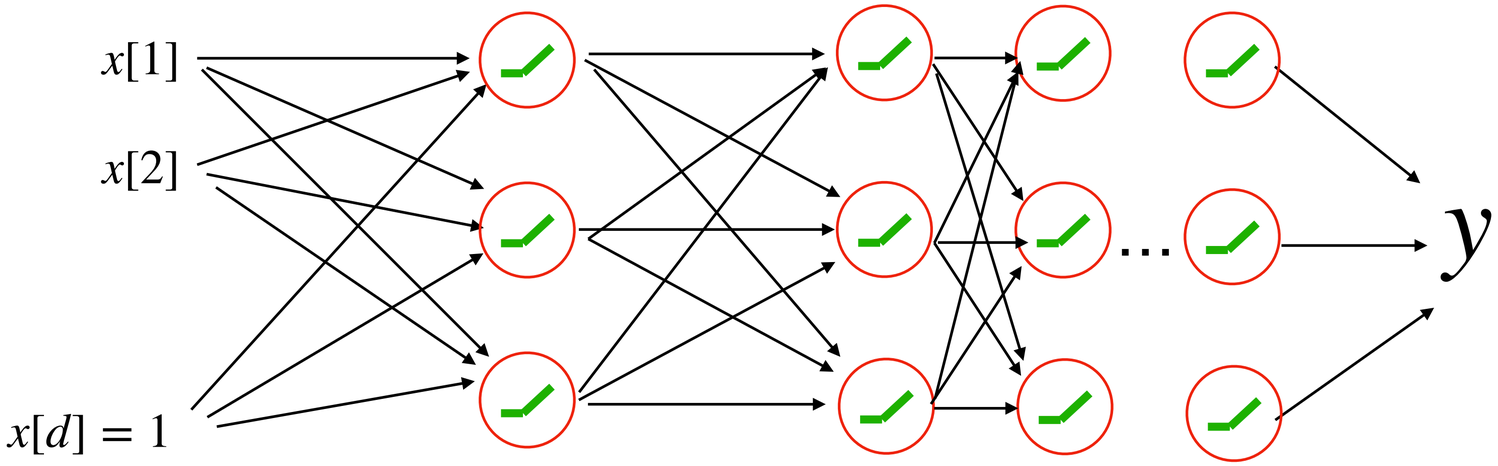

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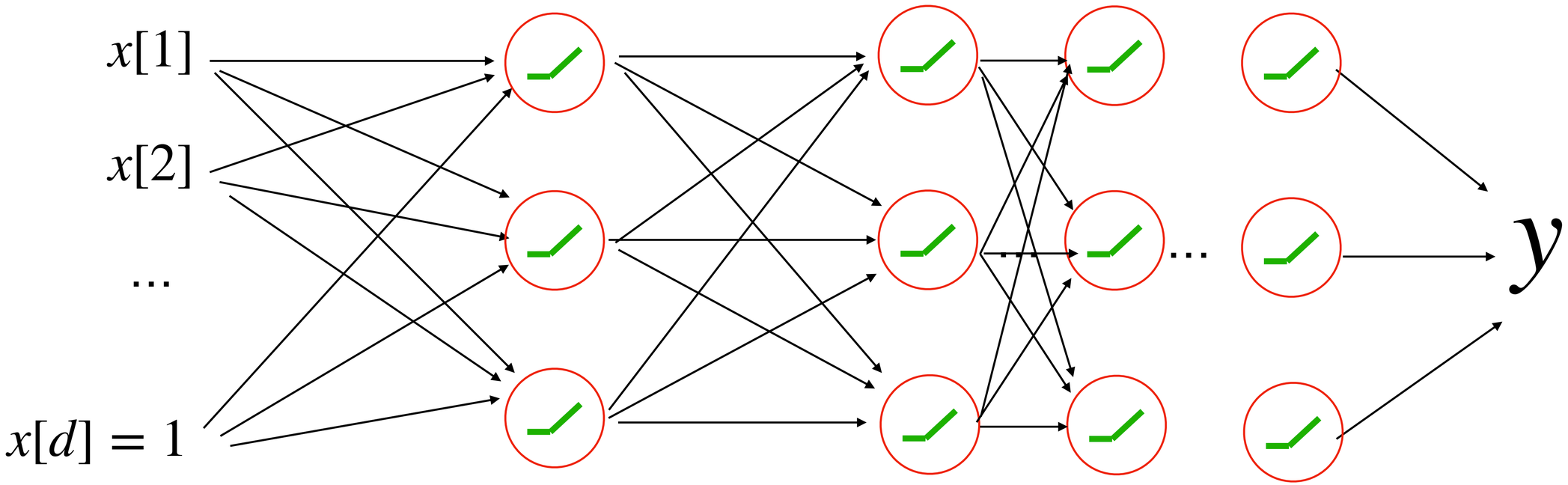
The benefits of going deep



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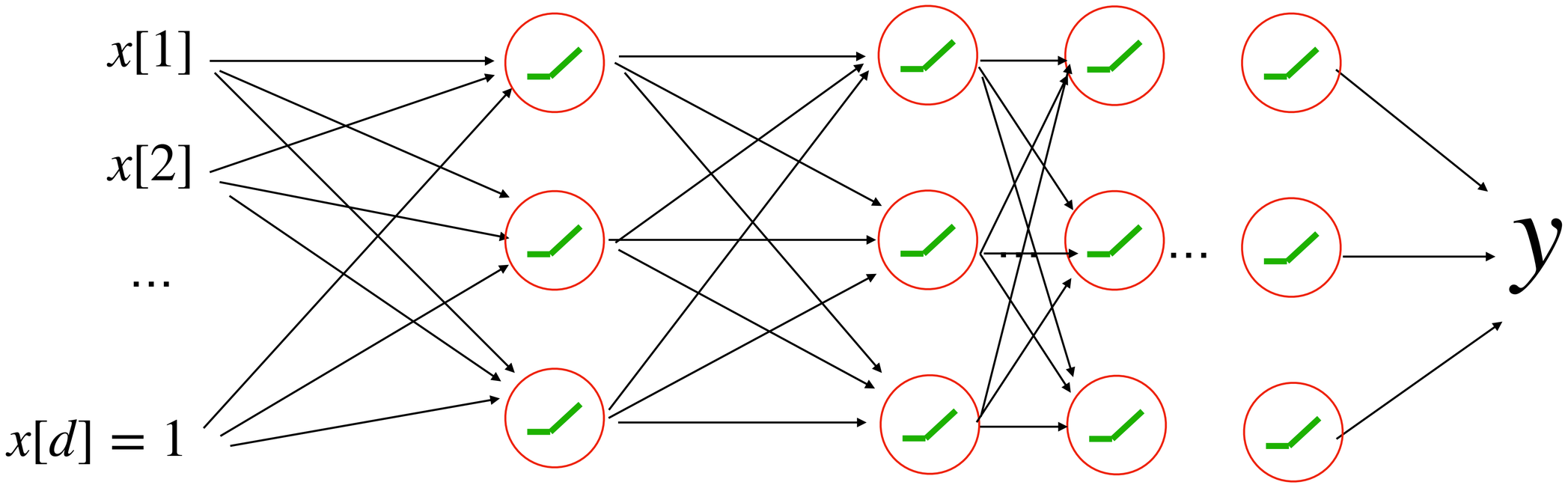


Allows us to represent complicated functions without making NN too wide

Outline of Today

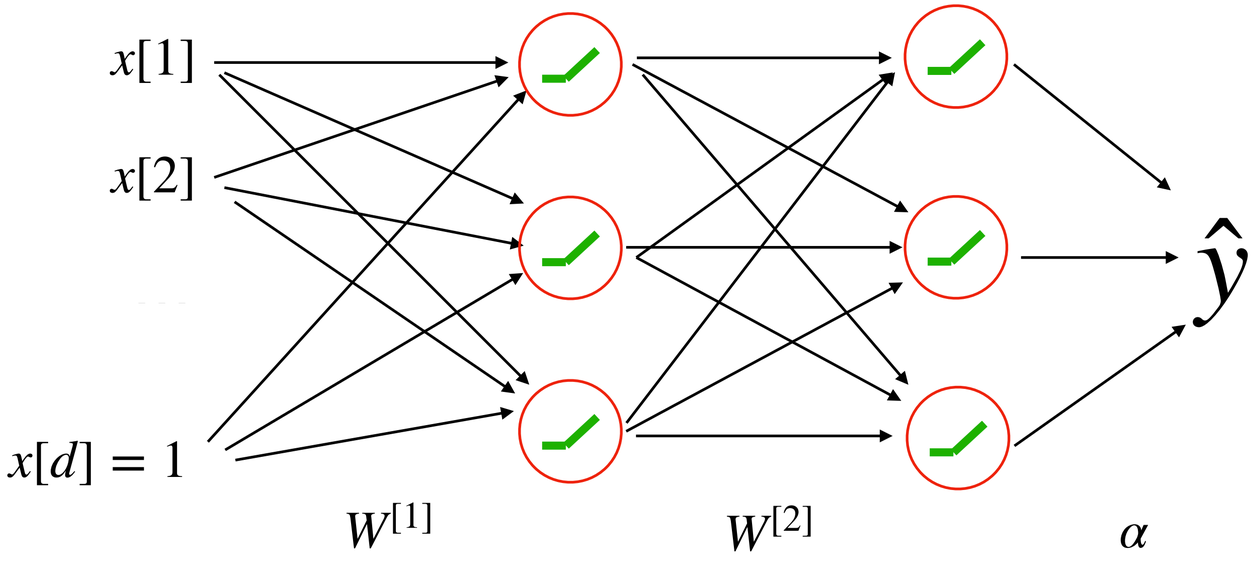
1. Analysis of Boosting

2. Multilayer feedforward Neural Network

3. Training a neural network

Training neural network via SGD

$$h(x) := \alpha^\top \text{ReLU}\left(W^{[2]}\text{ReLU}\left(W^{[1]}x\right)\right) + b$$



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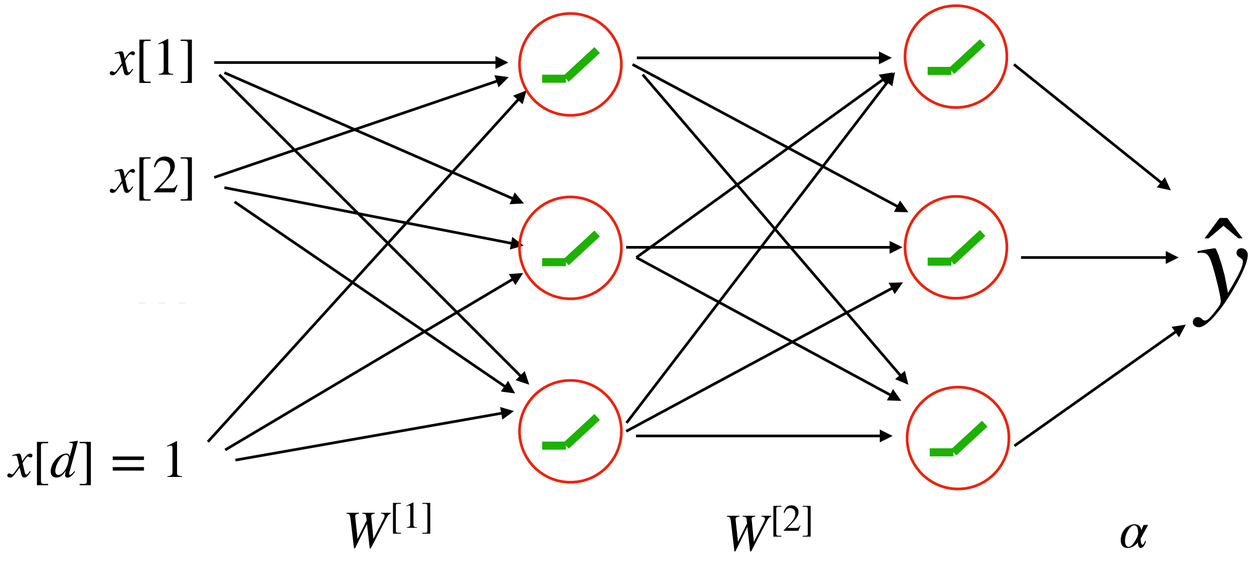

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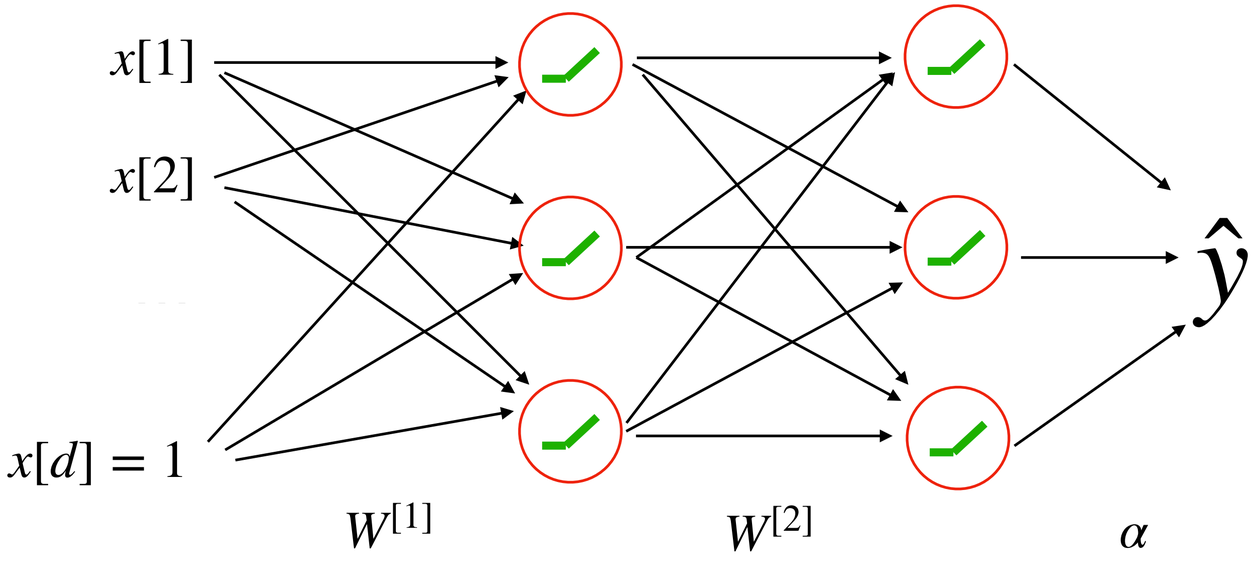

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Compute gradients:

$$\frac{\partial \ell(h(x), y)}{\partial W^{[1]}} \qquad \frac{\partial \ell(h(x), y)}{\partial W^{[2]}}$$

$$\frac{\partial \ell(h(x), y)}{\alpha} \qquad \frac{\partial \ell(h(x), y)}{b}$$

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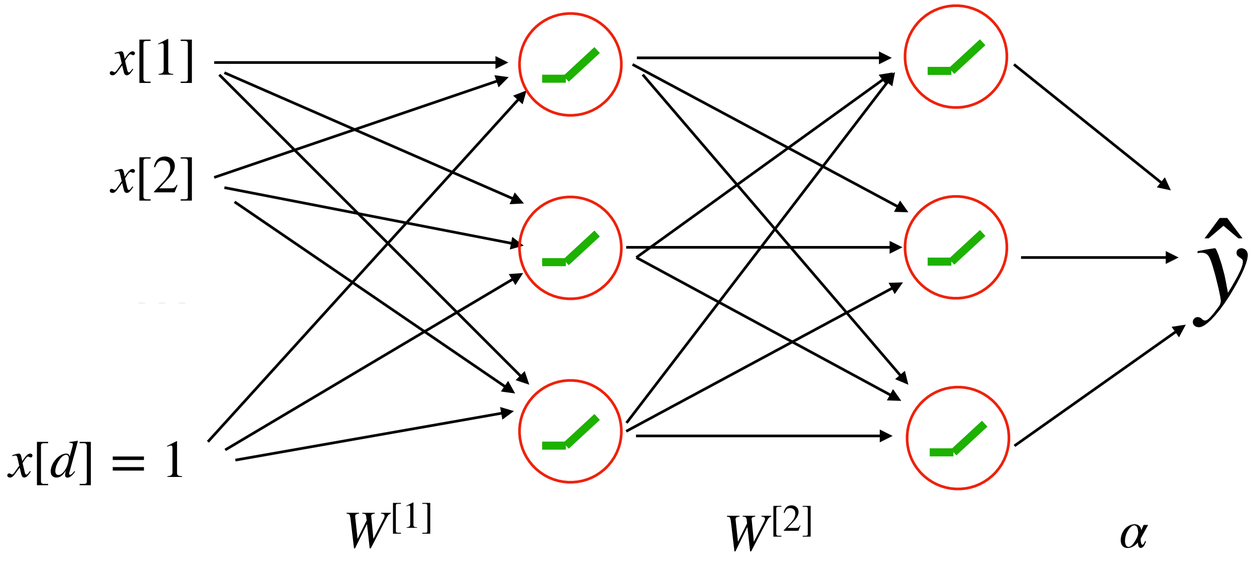

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(Next lecture: backpropagation for computing gradients)

Training neural network via SGD

Mini-batch Stochastic gradient descent

$\theta = [W^{[1]}, W^{[2]}, \alpha, b]$

For epoc $t = 1$ to T:

Training neural network via SGD

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// go through dataset multiple times

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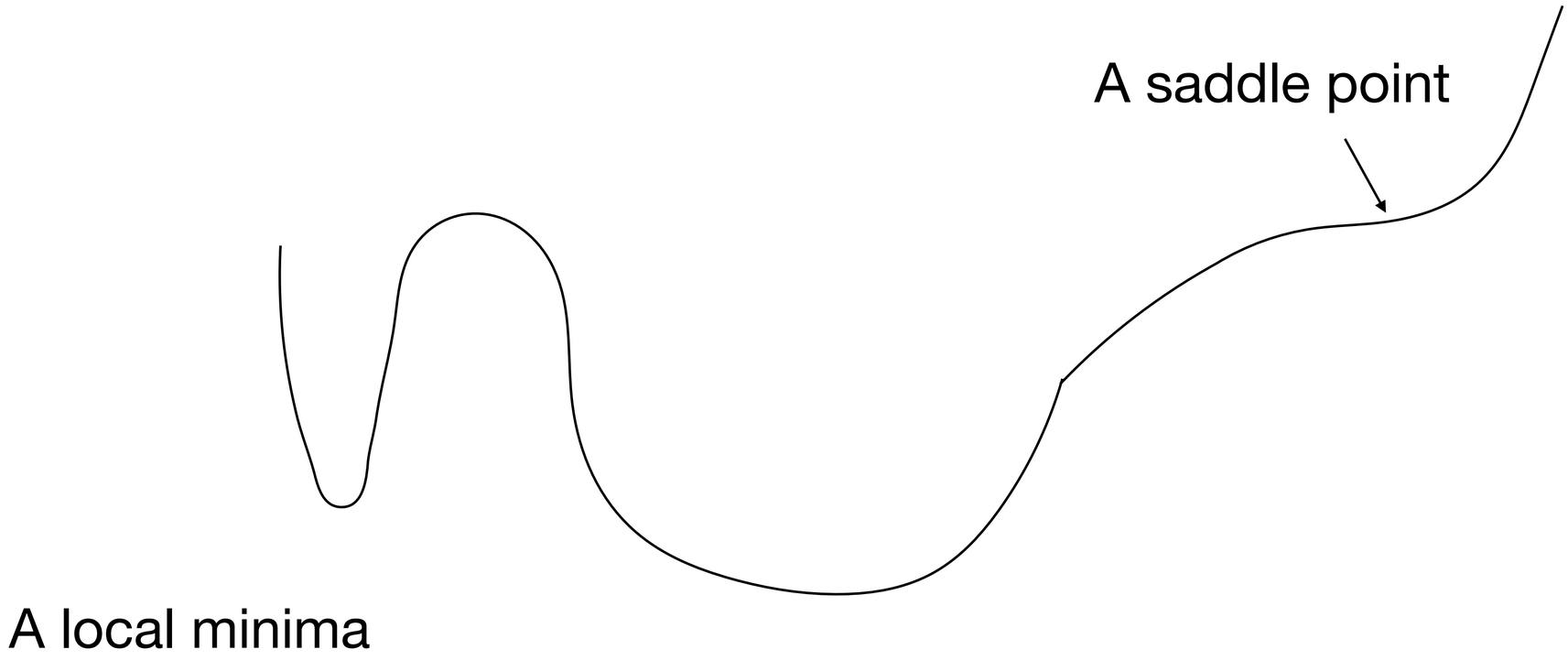
$\theta = \theta - \eta g$

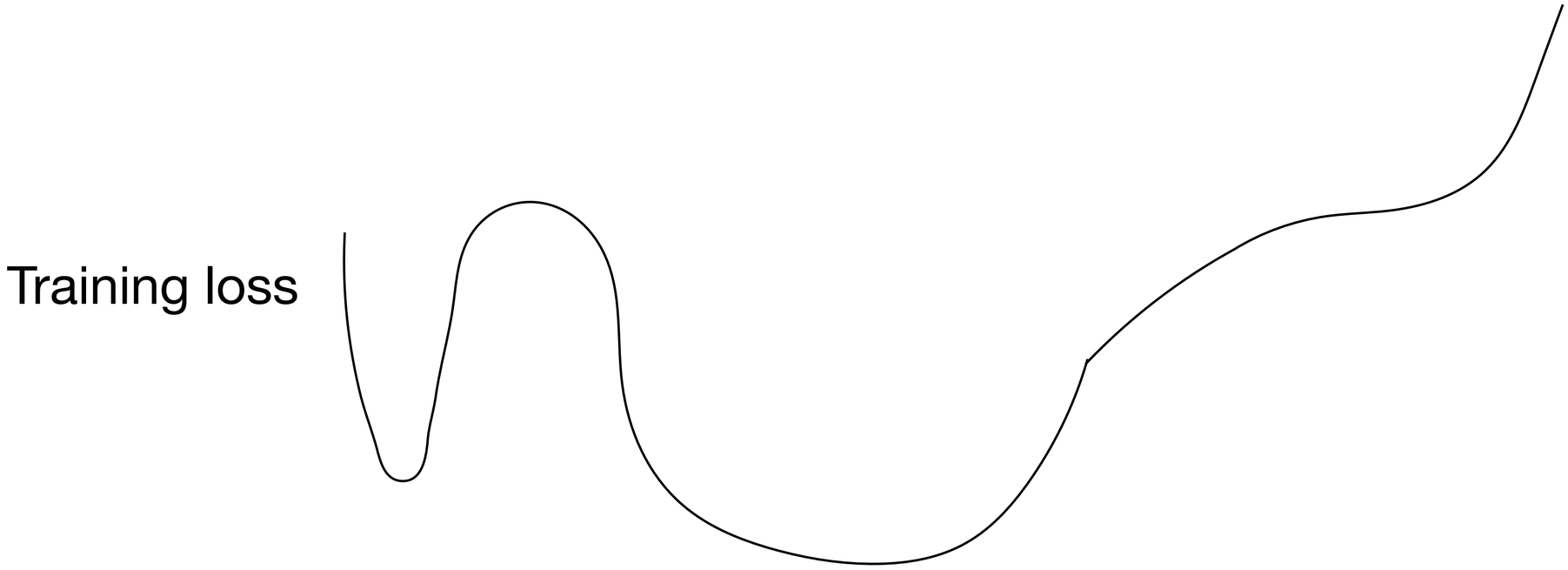
Training neural network via SGD

SGD helps avoiding local minima and saddle point



Training neural network via SGD

SGD tends to converge to a flat region



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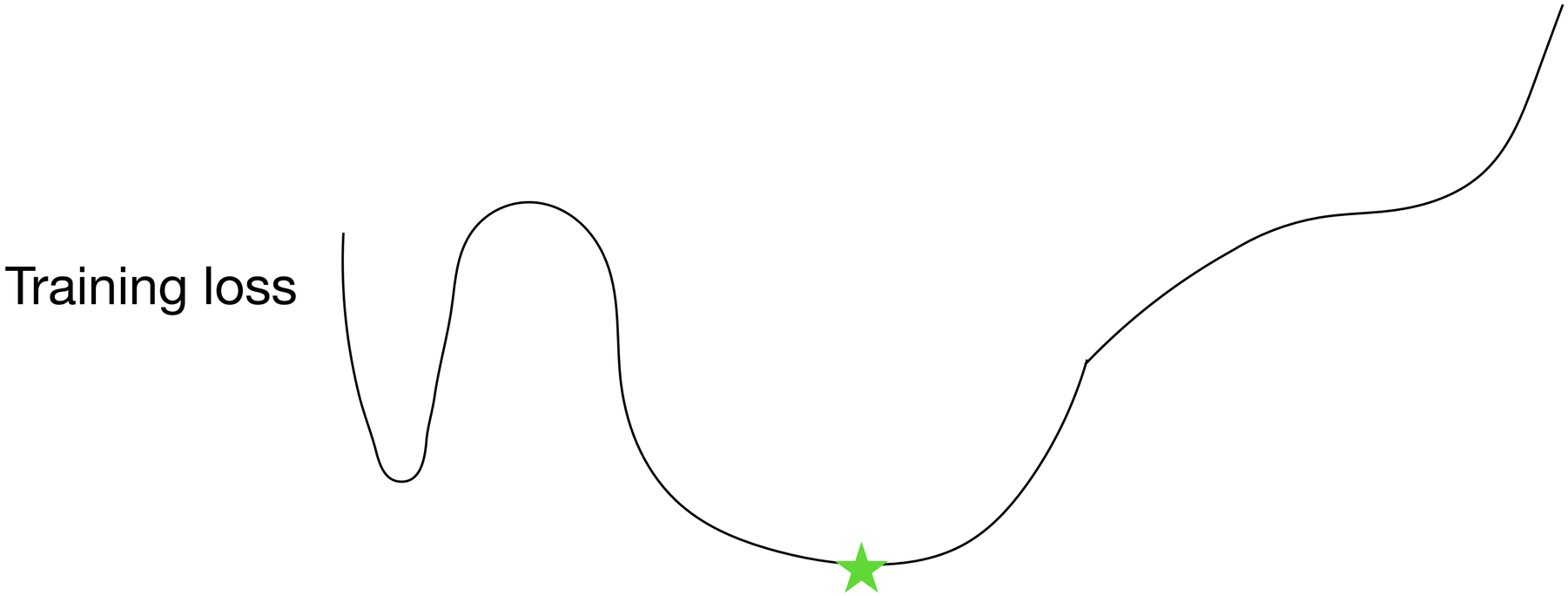

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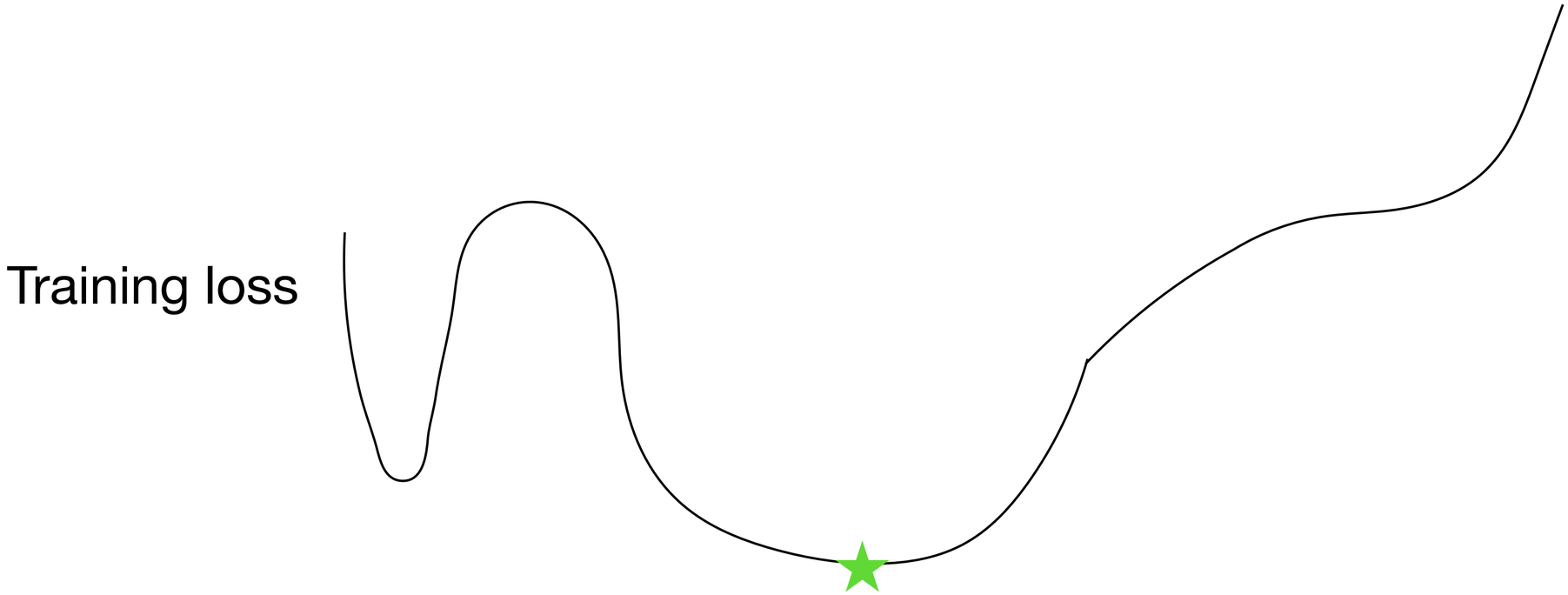

A flat local minima solution can help generalizes better to test data

Training neural network via SGD

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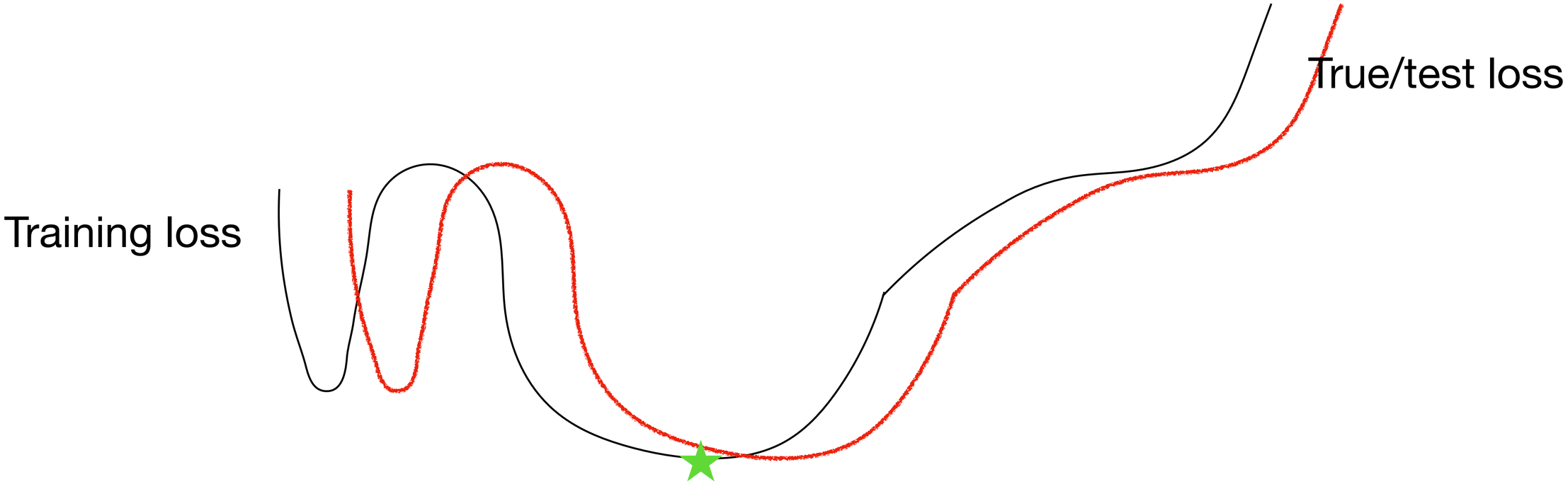

A flat local minima solution can help generalizes better to test data

Summary for today

1. Neural network is universal function approximation

2. SGD is important for training neural networks

Next lecture: backpropagation